

# THE COMPLEX OF CURVES PAIRWISE INTERSECTING AT MOST ONCE IS CONTRACTIBLE IN GENUS TWO

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ABSTRACT. Let  $S_2$  be the closed orientable surface of genus two and let  $C_1(S_2)$  be the flag complex on isotopy classes of essential simple closed curves in which a finite set of classes spans a simplex when the classes pairwise have geometric intersection number at most one. We prove that  $C_1(S_2)$  is contractible. The proof transports the whole problem to the six-times-marked sphere through the hyperelliptic involution, for which we give a self-contained proof of the required dictionary, and then runs a filtration spectral sequence on Hatcher’s contractible arc complex of the marked sphere. The single arithmetic input is that a genus two surface has exactly six Weierstrass points; it enters as a budget which forces every filtration stratum to have a contractible link. Along the way we prove that  $C_1(S_g)$  is simply connected for every  $g \geq 2$ , a statement asserted without proof by Souto; that part of the argument is not restricted to genus two.

## 1. INTRODUCTION

**1.1. The complexes  $C_d$ .** Let  $S_g$  be the closed connected orientable surface of genus  $g$  and let  $\mathcal{S}(S_g)$  denote the set of isotopy classes of essential simple closed curves in  $S_g$ ; *essential* means embedded and not bounding a disc. Write  $i(\cdot, \cdot)$  for the geometric intersection number, that is, the minimum of  $|a \cap b|$  over transverse representatives of the two classes. For an integer  $d \geq 0$  the  $d$ -curve complex  $C_d(S_g)$  is the simplicial complex with vertex set  $\mathcal{S}(S_g)$  in which a finite subset  $\{\gamma_0, \dots, \gamma_k\}$  spans a  $k$ -simplex if and only if  $i(\gamma_a, \gamma_b) \leq d$  for all  $a, b$ . Thus  $C_0(S_g)$  is the classical curve complex of Harvey, and each  $C_d(S_g)$  is by construction the flag complex on its own 1-skeleton. The inclusions  $C_0(S_g) \hookrightarrow C_1(S_g) \hookrightarrow \dots$  are equivariant for the action of the mapping class group.

The homotopy type of  $C_0(S_g)$  is known: by Harer [7] it is homotopy equivalent to a wedge of spheres of dimension  $2g - 2$ , so for  $g \geq 2$  it is simply connected — a fact we shall use in Section 7 — but it is never contractible. Souto [8] observed that the situation changes as soon as one allows curves to cross: he proved that the inclusion  $C_0(S) \hookrightarrow C_1(S)$  is homotopic to a constant map, and asked whether  $C_1(S)$  — or, if not, some  $C_d(S)$  — is contractible. That question is the subject of the present paper. Souto’s note is unpublished; the question is recorded as Question 2 of [8] in [9], and that is where we met it. It is genuinely a question about the ambient complex: all three of  $C_0(S_g)$ ,  $C_1(S_g)$  and  $C_d(S_g)$  have the same, infinite, vertex set, and  $C_1$  differs from  $C_0$  only by the additional simplices created by the edges with  $i = 1$ .

Two features of  $C_1$  make it harder to handle than  $C_0$ . First, it is not pure: it has maximal simplices of different dimensions, so no argument by collapsing top cells is available. Second, the natural sources of contractible models for curve complexes — arc complexes and their relatives — are complexes of *disjoint* objects, and the passage from “disjoint” to “crossing at most once” loses exactly the property that makes those models combinatorially transparent.

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**1.2. Results.** Our main theorem answers Souto's question affirmatively in genus two.

**Theorem 1.1.** *Let  $S_2$  be the closed orientable surface of genus two and  $C_1(S_2)$  the flag complex on isotopy classes of essential simple closed curves, with a simplex for each finite set of classes pairwise of geometric intersection number at most one. Then  $C_1(S_2)$  is contractible.*

The proof yields the homological statement first and the homotopy statement afterwards; the intermediate results are of independent interest and are stated separately.

**Theorem 1.2.**  $\tilde{H}_*(C_1(S_2); \mathbb{Z}) = 0$ .

**Theorem 1.3.** *For every  $g \geq 2$  the complex  $C_1(S_g)$  is simply connected.*

Note that Theorem 1.3 holds for *every*  $g \geq 2$ , not only for  $g = 2$ : it is the one statement in this paper which is not restricted to genus two, and its proof in Section 7 uses neither the hyperelliptic involution nor any of the genus two machinery. Theorem 1.3 is not new — it is asserted without proof by Souto [8] — but the argument we give in Section 7 is short and uses only Harer's determination of the homotopy type of  $C_0(S_g)$  together with an elementary surgery on an edge; in particular it does not use the contractibility statement for the arc-and-curve complex on which Souto's own route relies. Theorems 1.2 and 1.3 give Theorem 1.1 by Hurewicz and Whitehead.

Along the way we prove, in a self-contained manner, the genus two correspondence which is usually quoted as a consequence of the Birman–Hilden theorem. Let  $\iota$  be the hyperelliptic involution of  $S_2$ , let  $\pi: S_2 \rightarrow S_2/\iota \cong S^2$  be the quotient branched double cover and let  $B \subset S^2$  be its branch set, a set of six points. Write  $\Sigma = (S^2, B)$ .

**Theorem 1.4.** *The map  $\alpha \mapsto \pi^{-1}(\alpha)$  induces a bijection from the set of isotopy classes in  $\Sigma$  of essential arcs joining two distinct points of  $B$ , together with the set of isotopy classes of simple closed curves in  $S^2 \setminus B$  separating  $B$  into two sets of three, onto the set of isotopy classes of essential simple closed curves in  $S_2$ . Arcs correspond to non-separating curves and 3|3 circles to separating curves.*

Two further corollaries of Theorem 1.1 deserve mention. First,  $H_k(C_1(S_2); \mathbb{Z}) = 0$  for all  $k \geq 1$  and  $H_0 = \mathbb{Z}$ ; in particular the top reduced homology  $\tilde{H}_{11}$  vanishes, 11 being the dimension of  $C_1(S_2)$  (Proposition 6.2). Second, the full subcomplex of  $C_1(S_2)$  spanned by the non-separating classes is contractible as well.

**1.3. Method.** The proof has four movements.

(i) *Transport to the marked sphere.* A genus two surface carries a unique hyperelliptic involution  $\iota$ , with six fixed points, and  $\iota$  acts trivially on isotopy classes of simple closed curves. Consequently the whole complex  $C_1(S_2)$  can be read on the quotient orbifold  $\Sigma = (S^2, B)$ , where non-separating curves become essential arcs joining two distinct branch points and separating curves become circles splitting  $B$  as 3|3. We prove this dictionary rather than quote it (Section 4); the proof consists of a lifting lemma for ambient isotopies, a projection lemma for  $\iota$ -invariant curves, and a descent lemma whose argument goes back to Winarski's combinatorial proof of the Birman–Hilden theorem.

(ii) *The intersection formula.* Under the dictionary, the geometric intersection number upstairs is computed downstairs by

$$i(\tilde{\alpha}, \tilde{\beta}) = 2\hat{j}(\alpha, \beta) + e(\alpha, \beta),$$

where  $\hat{j}$  is the minimal number of interior intersections of the two arcs and  $e \in \{0, 1, 2\}$  the number of shared endpoints. The proof of this formula is where the choice of metric matters, and it is the step at which a naive argument fails. If one equips  $S^2 \setminus B$  with a complete hyperbolic

metric of finite area, then the six branch points become cusps, their preimages upstairs become cusps as well, and the preimage of an arc is *not* a closed geodesic; the standard bigon technology is then unavailable upstairs. Instead we put on  $(S^2, B)$  a hyperbolic *cone* metric with cone angle  $\pi$  at each of the six points. Its pullback under  $\pi$  has cone angle  $2\pi$  at each Weierstrass point, that is, it is a smooth closed hyperbolic metric on  $S_2$ , for which  $\iota$  is an isometry; preimages of arcs are then honest closed geodesics and the classical statement that distinct simple closed geodesics

bound obtained from the cone metric against a lower bound which is a purely topological count over the covering, once a perturbation lemma has been proved that makes the two minima range over the same set of representatives.

(iii) *A spectral sequence on the arc complex.* Let  $\mathcal{A} = A(S^2, B)$  be Hatcher's arc complex, which is contractible. Its simplices are collections of arcs with disjoint interiors;  $X$ , the complex which models the non-separating part of  $C_1(S_2)$ , is the subcomplex obtained by discarding two kinds of configuration: single loop arcs, and pairs of distinct arcs with the same pair of endpoints. Filtering  $\mathcal{A}$  by the number of vertices involved in such bad configurations produces a spectral sequence of a bounded filtration with augmentation — the row  $q = -1$  is present, so it is not first quadrant — whose  $E^1$ -page is  $\bigoplus_S \tilde{H}_q(L_S)$  over the strata  $S$ , converging to  $\tilde{H}_*(\mathcal{A}) = 0$  and whose column  $p = 0$  is  $\tilde{H}_*(X)$ . The whole work then consists in showing that the links  $L_S$  of the positive strata are contractible.

(iv) *A budget.* The link  $L_S$  of a stratum is a full flag subcomplex of  $X$ , and to prove it a cone it suffices to exhibit a vertex adjacent to all others. The stratum  $S$  decomposes canonically into *objects*: its loops, and its maximal families of arcs sharing a pair of endpoints. Each object is a connected embedded graph in  $S^2$  with at least two complementary discs, and we show that at least two of these complementary discs are *free*, meaning that no other object touches their closure. Each free disc must contain a marked point not touched by  $S$ , by essentiality; and the number of untouched marked points is  $6 - V'$  where  $V'$  is the number of marked points met by  $S$ . The count

$$k(g_1) + k(g_2) \leq 6 - V' \leq 5$$

forces one of the two free discs to contain exactly one or two marked points, and on such a disc an explicit cone point can be produced. This inequality is the only place where the number six enters, and it is the reason the argument stops at genus two.

**1.4. What is not claimed.** Three boundaries of the result should be stated explicitly.

First, *only  $g = 2$  is settled.* The identity  $6 = 2g + 2$  is a hard input to the budget argument of Section 9: at genus three there are eight branch points, the inequality above becomes vacuous, and the argument does not survive. More fundamentally, a general surface of genus  $g \geq 3$  carries no hyperelliptic involution at all, so the entire transport to a marked sphere is unavailable. **The method gives no information whatsoever about  $C_1(S_g)$  for  $g \geq 3$ .**

Second, *we prove nothing about the punctured versions, in either direction.* The natural simplicial map goes from the punctured complex to the closed one, by forgetting the puncture; it is surjective on vertices and not injective, since non-isotopic curves in  $\Sigma_{g,1}$  can become isotopic in  $S_g$ . We do not know whether  $C_1(\Sigma_{g,1})$  and  $C_1(S_g)$  are homotopy equivalent, and no argument here supplies an implication from either case to the other. In particular a reader should not take Theorem 1.1 to settle the punctured genus two case, and should not expect a proof for punctured or bordered surfaces to yield the closed case.

Third, *this paper contains no finite truncation computation.* All arguments are structural. We record this because homology computed on a finite pool of vertices, or on a finite quotient of such a pool, carries no information about the ambient complex: there is no map in either

direction relating the homology of a finite subcomplex spanned by a chosen finite set of curves to the homology of  $C_1(S_2)$ , and a non-vanishing Betti number of such a truncation is not evidence against Theorem 1.1.

**1.5. Organisation.** Section 2 fixes the complexes, the notion of isotopy that will be used throughout, and the transversality convention. Section 3 constructs the two hyperbolic structures and proves what is needed about geodesic representatives of arcs. Section 4 proves the hyperelliptic dictionary, Theorem 1.4. Section 5 proves the intersection number formula and deduces the adjacency criterion. Section 6 identifies  $C_1(S_2)$  up to homotopy with the complex  $\mathbf{X}$  on the marked sphere and computes its dimension. Section 7 proves Theorem 1.3. Section 8 sets up the spectral sequence on the augmented chain complex. Section 9 proves the budget lemma and produces a good face, and Section 10 proves that the links are cones. Section 11 assembles the proof of Theorems 1.2 and 1.1.

## 2. COMPLEXES, ARCS AND CONVENTIONS

**2.1. The surface and its involution.** Throughout,  $S_2$  is a closed connected orientable surface of genus two. A *curve* in  $S_2$  means an embedded circle; it is *essential* if it does not bound a disc. We write  $\mathcal{S}$  for the set of isotopy classes of essential curves, and

$$C_1(S_2) := \{ \{\gamma_0, \dots, \gamma_k\} \subset \mathcal{S} \text{ finite} : i(\gamma_a, \gamma_b) \leq 1 \text{ for all } a, b \},$$

the flag complex on the graph with vertex set  $\mathcal{S}$  and an edge whenever  $i \leq 1$ . We write  $C_1^{\text{ns}}(S_2)$  for the full subcomplex spanned by the classes of non-separating curves.

We fix once and for all the following model. Let  $B \subset S^2$  be a set of six points and let

$$\theta: \pi_1(S^2 \setminus B) \longrightarrow \mathbb{Z}/2$$

be the homomorphism sending each small meridian loop around a point of  $B$  to the non-trivial element;  $\theta$  is well defined because the product of the six meridians is trivial in  $\pi_1(S^2 \setminus B)$  and 6 is even, and it is surjective. Let  $S_2^\circ \rightarrow S^2 \setminus B$  be the connected double cover corresponding to  $\ker \theta$  and let

$$\pi: S_2 \longrightarrow S^2$$

be the branched double cover obtained by filling in one point over each point of  $B$ ; near such a point  $\pi$  has the local model  $z \mapsto z^2$ . Riemann–Hurwitz gives  $\chi(S_2) = 2 \cdot 2 - 6 = -2$ , so the total space is indeed a closed orientable surface of genus two, and every closed orientable surface of genus two is homeomorphic to it. We write

$$W := \pi^{-1}(B) \quad (\text{the six Weierstrass points}), \quad \iota: S_2 \rightarrow S_2$$

for the deck involution of the cover, so that  $\text{Fix}(\iota) = W$ ,  $\pi \circ \iota = \pi$ , and  $\pi$  restricts to an unbranched double covering  $\pi^\circ: S_2 \setminus W \rightarrow S^2 \setminus B$  with deck group  $\langle \iota \rangle$ . Finally set

$$\Sigma := (S^2, B).$$

Since all closed genus two surfaces are homeomorphic, it suffices to prove Theorem 1.1 for this model.

**2.2. Isotopy in the marked sphere.** Objects on  $\Sigma$  will always be compared by ambient isotopy of the sphere which does not move the marked points.

**Convention 2.1** (Isotopy in  $\Sigma$ ). Two subsets  $Z, Z' \subset S^2$  are *isotopic in  $\Sigma$* , written  $Z \simeq Z'$ , if there is an isotopy  $\{h_t\}_{t \in [0,1]}$  of  $S^2$  with  $h_0 = \text{id}$ , with  $h_t|_B = \text{id}$  for every  $t$ , and with  $h_1(Z) = Z'$ .

*Remark 2.2.* It is enough to require that each  $h_t$  preserve  $B$  setwise. Indeed  $B$  is finite hence discrete,  $t \mapsto h_t(b)$  is continuous with values in  $B$ , and  $h_0(b) = b$ ; so  $h_t(b) = b$  for all  $t$ .

Convention 2.1 is *not* isotopy relative to the boundary of a sub-disc, and it is *not* isotopy of  $S^2 \setminus B$  with the marked points deleted in a way that would freeze the direction in which an arc leaves a marked point: an arc may rotate freely about its endpoints. This will matter in Sections 5 and 10, where the results are sensitive to which notion is used, and we shall say so at the places concerned.

**2.3. Arcs.** We use Hatcher's notion of essential arc.

**Definition 2.3.** An *arc* in  $\Sigma$  is an embedded arc  $\alpha \subset S^2$  meeting  $B$  exactly in its two endpoints, which lie in  $B$  and are permitted to coincide. The arc is *essential* if it does not separate  $S^2$  into two components one of which is a disc meeting  $B$  only in the endpoints of  $\alpha$ . An arc whose endpoints coincide is a *loop*; otherwise it is a *non-loop arc*, and its *endpoint pair*  $\text{ep}(\alpha) \subset B$  is the two-element set of its endpoints.

**Observation 2.4.** Every non-loop arc is essential. Indeed if  $\alpha$  has distinct endpoints then  $S^2 \setminus \alpha$  is homeomorphic to an open disc, so  $\alpha$  does not separate  $S^2$  at all. Only loops can fail to be essential.

We also use  $3|3$  *circles*: embedded circles in  $S^2 \setminus B$  which separate  $B$  into two sets of three points.

**Definition 2.5.**  $\mathcal{A} := A(S^2, B)$  is the simplicial complex whose vertices are the isotopy classes (Convention 2.1) of essential arcs in  $\Sigma$ , loops included, and in which  $\{\alpha_0, \dots, \alpha_k\}$  spans a  $k$ -simplex when the classes admit representatives which are pairwise disjoint except possibly at endpoints and which are pairwise non-isotopic.

$X$  is the flag complex on the graph  $G$  whose vertices are the isotopy classes of *non-loop* arcs in  $\Sigma$ , with  $\alpha \sim \beta$  (for  $\alpha \neq \beta$ ) if and only if the two classes can be realised with disjoint interiors and  $\text{ep}(\alpha) \neq \text{ep}(\beta)$ .

*Remark 2.6.*  $X$  is a subcomplex of  $\mathcal{A}$ , but it is *not* a full subcomplex: two distinct non-loop arcs with the same endpoint pair can be realised with disjoint interiors, hence are adjacent in  $\mathcal{A}$ , while they are not adjacent in  $X$ .

**2.4. Transversality, smoothness, and the three intersection quantities.** Fix a smooth structure on  $S^2$  and give  $S_2$  the smooth structure for which  $\pi$  is smooth and has the local model  $z \mapsto z^2$  at each Weierstrass point; this is produced concretely in Section 3.

**Convention 2.7** (Smooth representatives). All representatives of arc classes are smooth embedded arcs which meet  $B$  only at their endpoints and have non-vanishing derivative at the endpoints.

**Convention 2.8** (Transversality). Two arcs  $\alpha, \beta$  are *transverse*, written  $\alpha \pitchfork \beta$ , if

- (a)  $\alpha \cap \beta$  is a *finite* set, and
- (b) at each point of  $\text{int } \alpha \cap \text{int } \beta$  the two tangent lines are distinct.

No condition is imposed at a shared endpoint.

The finiteness clause (a) is not implied by (b) and must be imposed. Indeed, in a coordinate  $u$  centred at a marked point  $b$ , put  $\alpha : u = t$  and  $\beta : u = t e^{ig(t)}$  with  $g(t) = e^{-1/t} \sin(1/t)$  for  $t > 0$  and  $g(0) = 0$ . Then  $g$  is smooth with all derivatives vanishing at 0, both arcs are smooth and regular at  $b$ , every interior intersection is transverse in the sense of (b), and the intersection set accumulates at  $b$ . Imposing (a) changes none of the minima below, since these minima are attained on pairs with finitely many intersections.

**Definition 2.9.** For transverse arcs  $\alpha, \beta$  set

$$j(\alpha, \beta) := |\text{int } \alpha \cap \text{int } \beta|, \quad e(\alpha, \beta) := \#\{\text{shared endpoints}\} \in \{0, 1, 2\},$$

and for isotopy classes  $a \neq b$  of essential arcs

$$\hat{j}(a, b) := \min\{j(\alpha, \beta) : \alpha \in a, \beta \in b, \alpha \pitchfork \beta\}.$$

**Lemma 2.10.** *e depends only on the pair of isotopy classes; explicitly, for non-loop arcs*

$$e(a, b) = |\text{ep}(a) \cap \text{ep}(b)|.$$

Moreover  $\pi^{-1}$  carries the set of shared endpoints bijectively onto  $\pi^{-1}(\text{ep}(a)) \cap \pi^{-1}(\text{ep}(b)) \subset W$ .

*Proof.* By Convention 2.1 an isotopy fixes  $B$  pointwise, and the endpoints of an arc lie in  $B$ ; hence the endpoint pair of an arc is unchanged along an isotopy, and  $\text{ep}$  is a function of the class. The set of shared endpoints is then  $\text{ep}(a) \cap \text{ep}(b)$ . Finally  $\pi|_W: W \rightarrow B$  is a bijection.  $\square$

By contrast  $j$  is *not* an isotopy invariant of a pair of classes; only its minimum  $\hat{j}$  is. Wherever the letter  $j$  appears below in a statement about classes it stands for  $\hat{j}$ .

### 3. TWO HYPERBOLIC STRUCTURES

The marked sphere  $\Sigma = (S^2, B)$  will carry two different hyperbolic structures, and it is essential to keep their roles apart. The *cone metric* of §3.1 has the six marked points as cone points of angle  $\pi$ ; its virtue is that its pullback to  $S_2$  is a *smooth* closed hyperbolic metric, so that preimages of arcs become honest closed geodesics. The *cusped metric* of §3.3 has the six marked points as cusps; its virtue is that arcs themselves — loops included — acquire complete geodesic representatives realising minimal intersection, simultaneously for an arbitrary family. Neither metric can do the other's work; see Remarks 3.7 and 4.3.

#### 3.1. A hyperbolic cone metric with all cone angles $\pi$ .

**Lemma 3.1.** *There is a regular geodesic hexagon in  $\mathbb{H}^2$  all of whose interior angles equal  $\pi/2$ .*

*Proof.* For  $t > 0$  let  $P(t) \subset \mathbb{H}^2$  be the hexagon whose vertices are the six points at distance  $t$  from a fixed origin along the geodesic rays making angles  $2\pi k/6$ ,  $k = 0, \dots, 5$ , consecutive vertices being joined by geodesic segments. The hexagon  $P(t)$  is embedded and its six interior angles are equal by symmetry; write  $\vartheta(t)$  for the common value. The vertices depend continuously on  $t$  and so does  $\vartheta$ .

As  $t \rightarrow 0^+$  the rescaled hexagons converge to the Euclidean regular hexagon, whose interior angle is  $2\pi/3$ ; hence  $\vartheta(t) \rightarrow 2\pi/3$ . As  $t \rightarrow \infty$  the vertices converge to six distinct points of  $\partial_\infty \mathbb{H}^2$  and  $P(t)$  converges, edge by edge, to the ideal regular hexagon, whose interior angles are 0; hence  $\vartheta(t) \rightarrow 0$ . Since  $\pi/2 \in (0, 2\pi/3)$ , the intermediate value theorem provides  $t_0 > 0$  with  $\vartheta(t_0) = \pi/2$ .  $\square$

We deliberately do not record the value of  $t_0$ : the construction below uses only the existence supplied by Lemma 3.1, and no numerical quantity enters any argument in this paper.

Write  $P := P(t_0)$ . By the Gauss–Bonnet formula for hyperbolic polygons, a geodesic  $n$ -gon with interior angles  $\theta_1, \dots, \theta_n$  (ideal vertices counted with  $\theta = 0$ ) has area  $(n - 2)\pi - \sum_i \theta_i$ ; thus

$$\text{Area}(P) = (6 - 2)\pi - 6 \cdot \frac{\pi}{2} = \pi. \tag{1}$$



**Proposition 3.2.** *Let  $D(P)$  be the double of  $P$ , that is, two isometric copies  $P^+, P^-$  of  $P$  glued by the identity along corresponding edges. Then  $D(P)$  is homeomorphic to  $S^2$  and carries a hyperbolic cone metric: it has constant curvature  $-1$  away from the six images of the vertices, and at each of these six points it has a cone point of angle  $\pi$ . Its total area is  $2\pi$ .*

*Proof.* Doubling a disc along its boundary gives a sphere. Away from the vertices, the metric on  $D(P)$  is smooth: in the interior of  $P^\pm$  this is clear, and along the interior of an edge the two sides are exchanged by the reflection which is an isometry of the local model, so the two hyperbolic half-discs glue to a hyperbolic disc. At an image of a vertex, two corners of angle  $\pi/2$  meet, one from  $P^+$  and one from  $P^-$ , giving a cone point of angle  $\pi$ . The area is  $2 \text{Area}(P) = 2\pi$  by (1). Consistently, the orbifold Euler characteristic of a sphere with six cone points of order two is  $\chi^{\text{orb}} = 2 - 6(1 - \frac{1}{2}) = -1$  and  $-2\pi\chi^{\text{orb}} = 2\pi$ .  $\square$

We identify the topological pair  $\Sigma = (S^2, B)$  with  $D(P)$  together with its six cone points, and write  $\bar{\rho}$  for the resulting hyperbolic cone metric on  $\Sigma$ . This is legitimate because any two six-element subsets of  $S^2$  are carried onto one another by a homeomorphism of  $S^2$ .

*Remark 3.3.* The metric  $\bar{\rho}$  determines a conformal, hence smooth, structure on  $S^2 \setminus B$ , and this structure extends over each cone point, a cone of angle  $\pi$  being conformally a disc. We take the smooth structure of Convention 2.7 to be this one. Since any two smooth structures on  $S^2$  are related by a diffeomorphism, and a diffeomorphism may be post-composed by one carrying any given six-element set to any other, the cusped metric of §3.3 may be, and is, transported to the same smooth manifold.

### 3.2. The pullback is a smooth closed hyperbolic surface.

**Proposition 3.4.** *Let  $\rho := \pi^*\bar{\rho}$ . Then  $\rho$  is a smooth hyperbolic metric on all of  $S_2$ , of area  $4\pi$ , and  $\iota$  is a  $\rho$ -isometry. In particular each Weierstrass point is a smooth point of  $(S_2, \rho)$ .*

*Proof.* On  $S_2 \setminus W$  the map  $\pi$  is a local homeomorphism, indeed a covering, so  $\rho$  is there a smooth metric of curvature  $-1$ . At  $w \in W$  the map  $\pi$  has the local model  $z \mapsto z^2$ , which pulls a cone point of angle  $\phi$  back to a cone point of angle  $2\phi$ ; here  $\phi = \pi$ , so the cone angle upstairs is  $2\pi$ , that is,  $w$  is a smooth point.

The following cell count verifies this directly. The two open hexagons in  $P^\pm$  are simply connected and disjoint from  $B$ , so each has two disjoint homeomorphic preimages;  $S_2$  is thus tiled by four hexagons. The interior of each of the six edges has two preimages, giving twelve edges, and the six cone points have six preimages, namely  $W$ . Hence

$$\chi(S_2) = 6 - 12 + 4 = -2,$$

as it must be. At each point of  $W$  exactly four hexagon corners meet, each of angle  $\pi/2$ , so the total angle is  $2\pi$ : a smooth point. The area is  $4 \text{Area}(P) = 4\pi = -2\pi\chi(S_2)$ .

Finally  $\pi \circ \iota = \pi$  gives  $\iota^*\rho = \iota^*\pi^*\bar{\rho} = (\pi \circ \iota)^*\bar{\rho} = \pi^*\bar{\rho} = \rho$ .  $\square$

Thus  $(S_2, \rho)$  is a closed hyperbolic surface. We record the three classical facts about closed hyperbolic surfaces that we shall use.

**Fact 3.5.** Let  $X$  be a closed hyperbolic surface.

- (G1) Every essential closed curve in  $X$  is freely homotopic to a unique closed geodesic, and if the curve is simple then that geodesic is simple.
- (G2) Distinct simple closed geodesics in  $X$  are in minimal position; in particular the number of their intersection points equals their geometric intersection number.
- (G3) Two essential simple closed curves in  $X$  are isotopic if and only if they are homotopic.

For (G1) see Farb–Margalit [2, Proposition 1.6]; the existence and uniqueness part alone is [2, Proposition 1.3], while the assertion that the geodesic representative of a simple curve is simple is the content of Proposition 1.6. Statement (G2) is [2, Corollary 1.9], and (G3) is [2, Proposition 1.10].

The hyperbolic metric  $\rho$  makes  $S_2$  into a closed Riemann surface of genus two, and  $\iota$ , being an orientation preserving isometry, is a conformal involution with quotient of genus zero; that is,  $\iota$  is the hyperelliptic involution of that Riemann surface. We may therefore invoke:

**Fact 3.6** (Haas–Susskind [5, Theorem 1]). Let  $J$  be the hyperelliptic involution of a genus two Riemann surface  $M$ . Then every simple closed geodesic on  $M$  is mapped onto itself by  $J$ . Moreover, if  $a$  is a simple closed geodesic, then  $J$  preserves the orientation of  $a$  when  $a$  is a dividing curve and reverses it when  $a$  does not divide  $M$ .

*Remark 3.7.* The cone angle  $\pi$  is smaller than  $2\pi$ , so  $(\Sigma, \bar{\rho})$  is *not* locally CAT(0) at the marked points: a path through a cone point can always be shortened by cutting the corner, and consequently  $(\Sigma, \bar{\rho})$  contains no geodesic passing through a marked point, and the usual existence and uniqueness statements for geodesic representatives are unavailable downstairs. We never use them there. In this paper the cone metric is used only through its pullback  $\rho$ , which lives on a smooth closed surface where Fact 3.5 applies verbatim; downstairs  $\bar{\rho}$  plays a purely topological role.

**3.3. The cusped metric and geodesic representatives of arcs.** Let  $P_\infty \subset \mathbb{H}^2$  be the ideal regular hexagon and let  $\Sigma^\circ$  be the double  $D(P_\infty)$  with its six ideal vertices removed. Doubling gives a complete hyperbolic metric  $h$  of curvature  $-1$ , smooth everywhere (the argument of Proposition 3.2 applies along the edges), with six cusps and area  $2 \cdot 4\pi = 8\pi = -2\pi\chi(\Sigma^\circ)$ , since  $\chi(S^2 \setminus B) = 2 - 6 = -4$ . Identifying  $D(P_\infty)$  with  $(S^2, B)$  as in Remark 3.3, we regard  $h$  as a complete hyperbolic metric of finite area on  $\Sigma^\circ := S^2 \setminus B$ .

We keep the following notation for the rest of §3.3. Let  $p: \mathbb{H}^2 \rightarrow \Sigma^\circ$  be the universal cover with deck group  $\Gamma \cong \pi_1(\Sigma^\circ)$ , a free group of rank five;  $\Gamma$  is discrete, acts freely, and is torsion free. For each  $b \in B$  fix an embedded cusp neighbourhood  $C_b \subset \Sigma^\circ$ , the  $C_b$  pairwise disjoint, so that  $p^{-1}(C_b)$  is a disjoint union of open horoballs, any two of which are equal or disjoint; each such horoball  $H$  has a base point in  $\partial_\infty \mathbb{H}^2$  whose stabiliser in  $\Gamma$  is infinite cyclic generated by a parabolic, and  $H/\text{Stab}(H) \cong C_b$ . Let  $P_\infty^\Gamma \subset \partial_\infty \mathbb{H}^2$  denote the set of base points of all these horoballs. If  $\alpha$  is an essential arc, we write  $\alpha^\circ := \alpha \setminus B$ , a properly embedded line in  $\Sigma^\circ$ . Finally we write  $m_b \in H_1(\Sigma^\circ; \mathbb{Z})$  for the class of a small loop around  $b$ ; one has  $H_1(\Sigma^\circ; \mathbb{Z}) \cong \mathbb{Z}^B / (\sum_{b \in B} m_b)$ .

After shrinking the  $C_b$  we may and do assume that for each of the finitely many arcs under consideration at a given moment,  $\alpha \cap C_b$  consists exactly of the ends of  $\alpha$  running into  $b$ .

**Lemma 3.8** (Ideal endpoints). *Let  $\alpha$  be an arc in  $\Sigma$  and  $\tilde{\alpha} \subset \mathbb{H}^2$  a lift of  $\alpha^\circ$ . Then each end of  $\tilde{\alpha}$  converges to a point of  $P_\infty^\Gamma$ , the two limits  $x, y$  being the base points of the horoballs containing the two ends, and  $\tilde{\alpha} \cup \{x, y\}$  is an arc in the closed disc  $\overline{\mathbb{H}^2} = \mathbb{H}^2 \cup \partial_\infty \mathbb{H}^2$ . We write  $\partial_\infty \tilde{\alpha} := \{x, y\}$ .*

*Proof.* One end of  $\alpha^\circ$  eventually lies in  $C_b$ ,  $b$  being the corresponding endpoint of  $\alpha$ ; that end is connected, so its lift eventually lies in a single component  $H_x$  of  $p^{-1}(C_b)$ , with base point  $x$ . The closure of  $H_x$  in  $\overline{\mathbb{H}^2}$  meets  $\partial_\infty \mathbb{H}^2$  only in  $x$ , and the end is a proper ray in  $\mathbb{H}^2$ , so it leaves every compact set and its limit set is contained in  $\{x\}$ ; hence it converges to  $x$ . The same applies to the other end, with limit  $y$ . Extending a parametrisation  $\mathbb{R} \rightarrow \mathbb{H}^2$  of  $\tilde{\alpha}$  to  $[-\infty, +\infty] \rightarrow \overline{\mathbb{H}^2}$  gives a continuous injection from a compact space to a Hausdorff space, hence a homeomorphism onto its image.  $\square$

**Lemma 3.9** (Essential arcs have distinct ideal endpoints). *If  $\alpha$  is essential then  $x \neq y$ .*



*Proof.* If the two endpoints of  $\alpha$  are distinct points  $b \neq b'$  then  $x$  and  $y$  are base points of horoballs lying over different cusps, hence in different  $\Gamma$ -orbits, so  $x \neq y$ .

Suppose then that  $\alpha$  is a loop based at  $b$ , and assume  $x = y$ ; write  $H := H_x$  and let  $\text{Stab}_\Gamma(x) = \langle g \rangle$ , generated by a parabolic, with  $g$  corresponding to the loop  $\partial C_b$ . The arc  $\alpha$  separates  $S^2$  into two open discs  $D_1, D_2$ ; put  $n_i := |D_i \cap B|$ , so  $n_1 + n_2 = 5$ , and essentiality says exactly that  $n_1 \geq 1$  and  $n_2 \geq 1$ .

The two ends of  $\alpha$  divide the circle  $\partial C_b$  into two arcs; let  $c_1$  be the one contained in  $D_1$  and let

$$\gamma_1 := (\alpha \setminus C_b) \cup c_1,$$

a simple closed curve in  $\Sigma^\circ$ . On one side of  $\gamma_1$  lie the  $n_1$  marked points of  $D_1 \cap B$ ; on the other side lie  $b$  and the  $n_2$  marked points of  $D_2 \cap B$ . Consequently, with a suitable orientation,

$$[\gamma_1] = \sum_{q \in S} m_q \in H_1(\Sigma^\circ; \mathbb{Z}), \quad S := D_1 \cap B, \quad b \notin S, \quad |S| = n_1. \quad (2)$$

We now compute  $[\gamma_1]$  a second way. Since  $x = y$ , both ends of  $\tilde{\alpha}$  lie in  $H$ ; as  $\alpha \cap C_b$  consists exactly of the two ends and  $\alpha$  is embedded,  $\tilde{\alpha} \cap p^{-1}(C_b)$  consists exactly of those two ends, so  $K := \tilde{\alpha} \setminus H$  is a compact arc with endpoints  $p_-, p_+ \in \partial H$ . Lift  $\gamma_1$  starting at  $p_-$ : the lift runs along  $K$  to  $p_+$  and then along a lift of  $c_1$ , which is a path in  $\partial H$ , ending at some  $q \in \partial H$ . If  $k \in \Gamma$  is the deck transformation corresponding to  $[\gamma_1]$  under this choice of base point, then  $q = k(p_-)$ . Now  $p_- \in \partial H$  and  $k(p_-) \in \partial H$  force  $k(H) \cap H \neq \emptyset$ , hence  $k(H) = H$  and  $k \in \langle g \rangle$ . Therefore  $[\gamma_1]$  is an integral multiple of  $m_b$  in  $H_1(\Sigma^\circ; \mathbb{Z})$ , say  $[\gamma_1] = \kappa m_b$ .

Comparing with (2) in  $\mathbb{Z}^B / (\sum_{q \in B} m_q)$  means that in  $\mathbb{Z}^B$

$$\mathbf{1}_S - \kappa \mathbf{e}_b = \lambda \mathbf{1}_B \quad \text{for some } \lambda \in \mathbb{Z}.$$

Reading off the coordinate at a point  $q \notin S \cup \{b\}$  gives  $\lambda = 0$ ; reading off the coordinate at a point  $q \in S$  gives  $\lambda = 1$ . Hence either  $S = \emptyset$ , that is  $n_1 = 0$ , or  $S \cup \{b\} = B$ , that is  $n_1 = 5$  and  $n_2 = 0$ . Both contradict essentiality.  $\square$

**Lemma 3.10** (Linking forces crossing). *Let  $\ell_1, \ell_2 \subset \mathbb{H}^2$  be properly embedded lines with ideal endpoint pairs  $\{x, y\}$  and  $\{u, v\}$ , the four points being distinct, and suppose  $\{u, v\}$  separates  $\{x, y\}$  on  $\partial_\infty \mathbb{H}^2$ . Then  $\ell_1 \cap \ell_2 \neq \emptyset$ .*

*Proof.* By Lemma 3.8,  $\bar{\ell}_1 := \ell_1 \cup \{x, y\}$  is an arc in the closed disc  $\overline{\mathbb{H}^2}$  with its endpoints on the boundary circle and its interior in the open disc. Let  $A_1, A_2$  be the two open arcs of  $\partial_\infty \mathbb{H}^2 \setminus \{x, y\}$ . Then  $\bar{\ell}_1 \cup \bar{A}_1$  is a Jordan curve, and by the Jordan curve theorem  $\overline{\mathbb{H}^2} \setminus \bar{\ell}_1$  has exactly two components  $U_1 \supseteq A_1$  and  $U_2 \supseteq A_2$ , both open in  $\overline{\mathbb{H}^2}$ . Linking means, after renaming,  $u \in A_1$  and  $v \in A_2$ . The set  $\bar{\ell}_2$  is connected and meets both  $U_1$  and  $U_2$ , hence meets  $\bar{\ell}_1$ ; since the four ideal points are distinct, the intersection point is not ideal.  $\square$

**Corollary 3.11.** *Two distinct complete geodesics of  $\mathbb{H}^2$  meet if and only if their ideal endpoint pairs link, and in that case they meet in exactly one point, transversally.*

*Proof.* If they link they meet, by Lemma 3.10. If they meet in two points they coincide, since a geodesic through two points of  $\mathbb{H}^2$  is unique; and if they shared an ideal endpoint or had unlinked endpoint pairs they would be disjoint. Meeting in exactly one point with distinct tangents is transversality.  $\square$

**Proposition 3.12** (Geodesic representatives of arcs). *Let  $a$  be an isotopy class of essential arcs in  $\Sigma$ ,  $\alpha_0 \in a$ , and let  $\{x, y\} = \partial_\infty \tilde{\alpha}_0$  for some lift  $\tilde{\alpha}_0$ ;  $x \neq y$  by Lemma 3.9. Let  $\ell$  be the complete geodesic of  $\mathbb{H}^2$  with ideal endpoints  $x, y$  and put  $a^* := p(\ell)$ . Then*

- (1)  $a^*$  is a complete geodesic of  $(\Sigma^\circ, h)$  whose closure in  $S^2$  is an arc of  $\Sigma$  isotopic to  $\alpha_0$ ;

- (2)  $a^*$  depends only on  $a$ , and is the unique complete geodesic representative in  $a$ ;
- (3)  $a^*$  is simple, that is  $p|_\ell$  is injective.

*Proof.* (i) Let  $r: \mathbb{H}^2 \rightarrow \ell$  be the nearest point projection; it is continuous. Put  $F(s, t) :=$  the point at parameter  $t$  on the geodesic segment  $[\tilde{\alpha}_0(s), r(\tilde{\alpha}_0(s))]$ . Then  $F$  is continuous,  $F(\cdot, 0) = \tilde{\alpha}_0$  and  $F(\cdot, 1) = r \circ \tilde{\alpha}_0$ , whose image is  $\ell$ . The homotopy is proper: as  $s \rightarrow -\infty$  both  $\tilde{\alpha}_0(s)$  and  $r(\tilde{\alpha}_0(s))$  converge to  $x$ , and horoballs are convex, so once both endpoints of a geodesic segment lie in a horoball based at  $x$  the whole segment does; hence  $F((-\infty, s_0] \times [0, 1])$  lies in that horoball for  $s_0$  small enough. Symmetrically at the other end. Projecting,  $p \circ F$  is a proper homotopy from  $\alpha_0^\circ$  to  $a^*$  whose two ends stay in the respective cusp neighbourhoods throughout. Adding the endpoints, we obtain a homotopy of arcs in  $\Sigma$  rel  $B$  between two *embedded* arcs. Here we use Epstein's theorem [10] that two embedded arcs in a surface which are homotopic rel their endpoints are ambient isotopic; it is the arc counterpart of Fact 3.5(G3) and is what upgrades the homotopy just produced to an isotopy in the sense of Convention 2.1. We flag the use explicitly rather than absorb it into the word “extends”. In particular the closure of  $a^*$  in  $S^2$  is an arc with the same endpoints as  $\alpha_0$ .

(ii) The ideal endpoint pair is invariant under proper homotopy of arcs. Indeed if  $H: [0, 1]^2 \rightarrow S^2$  is a homotopy through arcs with  $H(0, t) \equiv b$ , then by compactness there is  $\delta > 0$  with  $H((0, \delta) \times [0, 1]) \subset C_b$ ; a lift of this connected set lies in a single horoball, so the corresponding ends of all the lifted arcs converge to the same ideal point. The same holds at the other end. Two complete geodesics with the same ideal endpoint pair coincide, whence uniqueness.

(iii) First, the setwise stabiliser of  $\ell$  in  $\Gamma$  is trivial. An element  $\gamma$  with  $\gamma\ell = \ell$  permutes  $\{x, y\}$ ; if it swaps them then  $\gamma^2$  fixes both, and since  $\Gamma$  is torsion free and free,  $\gamma^2 = 1$  forces  $\gamma = 1$ ; if it fixes both then  $\gamma$  is hyperbolic with axis  $\ell$  or the identity, and a hyperbolic element cannot share exactly one fixed point with a parabolic element in a discrete group, while  $\text{Stab}_\Gamma(x)$  contains a parabolic. Hence  $\gamma = 1$ . The same argument applies to  $\tilde{\alpha}_0$ , whose ideal endpoint pair is also  $\{x, y\}$ .

Now suppose  $p|_\ell$  is not injective. Then there is  $\gamma \in \Gamma \setminus \{1\}$  with  $\gamma\ell \cap \ell \neq \emptyset$ , and  $\gamma\ell \neq \ell$  by the previous paragraph. By Corollary 3.11 the pairs  $\{x, y\}$  and  $\{\gamma x, \gamma y\}$  link. But  $\tilde{\alpha}_0$  and  $\gamma\tilde{\alpha}_0$  have exactly these ideal endpoint pairs, so they meet by Lemma 3.10. Taking  $z \in \gamma\tilde{\alpha}_0 \cap \tilde{\alpha}_0$ , the two distinct points  $\gamma^{-1}z$  and  $z$  of  $\tilde{\alpha}_0$  have the same image under  $p$ , contradicting the injectivity of  $p|_{\tilde{\alpha}_0}$ , which holds because  $\alpha_0$  is embedded.  $\square$

**Theorem 3.13** (Geodesic representatives realise  $\hat{j}$ , simultaneously). *Let  $a \neq b$  be isotopy classes of essential arcs in  $\Sigma$  with geodesic representatives  $a^*, b^*$ . Then for every pair of transverse representatives  $\alpha_0 \in a$ ,  $\beta_0 \in b$ ,*

$$j(a^*, b^*) \leq j(\alpha_0, \beta_0),$$

*so  $j(a^*, b^*) = \hat{j}(a, b)$ . Consequently, for any family  $a_1, \dots, a_k$  of pairwise distinct classes of essential arcs, the geodesic representatives  $a_1^*, \dots, a_k^*$  realise  $\hat{j}$  for every pair simultaneously; in particular, if  $\hat{j}(a_m, a_n) = 0$  for all  $m, n$ , then the  $a_m^*$  have pairwise disjoint interiors.*

*Proof.* The classes being distinct,  $a^* \neq b^*$  by Proposition 3.12(ii). Fix a lift  $\ell$  of  $a^*$  and let  $\tilde{\alpha}_0$  be the lift of  $\alpha_0^\circ$  with the same ideal endpoint pair, which exists by Proposition 3.12(ii). Index the lifts of  $b^*$  and of  $\beta_0^\circ$  by  $\Gamma$  compatibly, so that  $\tilde{\beta}^{(\gamma)}$  and  $\tilde{\beta}_0^{(\gamma)}$  have the same ideal endpoint pair for each  $\gamma$ ; this is possible because both families are free  $\Gamma$ -orbits and the ideal endpoint pair is  $\Gamma$ -equivariant.

Since  $a^*$  is simple,  $p|_\ell$  is injective and

$$j(a^*, b^*) = |\ell \cap p^{-1}(b^*)| = \sum_{\gamma} |\ell \cap \tilde{\beta}^{(\gamma)}|.$$

Each summand is at most 1 by Corollary 3.11, so with  $J := \#\{\gamma : \ell \cap \tilde{\beta}^{(\gamma)} \neq \emptyset\}$  we get  $j(a^*, b^*) = J$ . If  $\ell \cap \tilde{\beta}^{(\gamma)} \neq \emptyset$  then the two ideal endpoint pairs link by Corollary 3.11, hence  $\tilde{\alpha}_0$  and  $\tilde{\beta}_0^{(\gamma)}$  meet by Lemma 3.10. Therefore

$$j(\alpha_0, \beta_0) = |\tilde{\alpha}_0 \cap p^{-1}(\beta_0^\circ)| = \sum_{\gamma} |\tilde{\alpha}_0 \cap \tilde{\beta}_0^{(\gamma)}| \geq J.$$

This proves the inequality; taking the infimum over transverse pairs gives  $j(a^*, b^*) = \hat{j}(a, b)$ . As the geodesic representative of a class depends only on the class, a family of classes gives a fixed family of curves realising  $\hat{j}$  for every pair at once.  $\square$

Theorem 3.13 also shows, a posteriori, that geodesic representatives bound no empty bigon and no empty half-bigon: such a configuration could be removed by a homotopy supported near the corresponding disc, lowering  $j$  below  $\hat{j}$ . In the half-bigon case the homotopy rotates the arc about the marked point which is a corner of the disc, which is legitimate precisely because the isotopies of Convention 2.1 allow an arc to turn about its endpoints. We shall not need this remark, but it explains why no bigon criterion for arcs is invoked anywhere in this paper: Theorem 3.13 was proved directly in  $\mathbb{H}^2$ , using only that two distinct complete geodesics meet at most once.

#### 4. THE HYPERELLIPTIC DICTIONARY

This section proves Theorem 1.4. Nothing here is quoted from the Birman–Hilden theory in the form of a black box; the one argument we borrow, the descent lemma of §4.4, is reproduced in full with its attribution and with the step that our situation requires supplied.

Throughout,  $\mathcal{A}^\neq$  denotes the set of isotopy classes (Convention 2.1) of essential arcs of  $\Sigma$  with two *distinct* endpoints,  $\mathcal{C}$  the set of isotopy classes of 3|3 circles, and  $\mathcal{S}$  the set of isotopy classes of essential simple closed curves in the closed surface  $S_2$ . Recall  $\theta: \pi_1(S^2 \setminus B) \rightarrow \mathbb{Z}/2$  from §2: it sends every meridian of a point of  $B$  to the non-trivial element, and a loop lifts to a loop in  $S_2 \setminus W$  if and only if  $\theta$  kills it. We will use repeatedly the form of Riemann–Hurwitz appropriate here: if  $q: \tilde{Y} \rightarrow Y$  is a double cover of compact surfaces, branched over  $k$  interior points with local model  $z \mapsto z^2$ , then  $\chi(\tilde{Y}) = 2\chi(Y) - k$ .

##### 4.1. Lifting isotopies.

**Lemma 4.1.** *Let  $H: S^2 \times [0, 1] \rightarrow S^2$  be continuous, with each  $H_t := H(\cdot, t)$  a homeomorphism,  $H_0 = \text{id}$  and  $H_t(B) = B$  for all  $t$ . Then there is a unique continuous  $\tilde{H}: S_2 \times [0, 1] \rightarrow S_2$  with  $\tilde{H}_0 = \text{id}$  and  $\pi \circ \tilde{H}_t = H_t \circ \pi$  for all  $t$ . Each  $\tilde{H}_t$  is a homeomorphism commuting with  $\iota$  and fixing  $W$  pointwise. Consequently, for every  $\iota$ -invariant subset  $Z \subset S^2$ ,*

$$\tilde{H}_1(\pi^{-1}(Z)) = \pi^{-1}(H_1(Z)).$$

*Proof.* By Remark 2.2,  $H_t|_B = \text{id}$ , so  $H_t$  preserves  $X := S^2 \setminus B$  and  $F := H \circ (\pi^\circ \times \text{id}): (S_2 \setminus W) \times [0, 1] \rightarrow X$  is continuous. As  $\pi^\circ$  is a covering map, the homotopy lifting property and the uniqueness of lifts [3, Propositions 1.30 and 1.34] give a unique continuous lift  $\tilde{H}$  of  $F$  on  $(S_2 \setminus W) \times [0, 1]$  with  $\tilde{H}_0 = \text{id}$ ; it satisfies  $\pi \tilde{H}_t = H_t \pi$  there.

Lifting the inverse isotopy  $t \mapsto H_t^{-1}$  likewise gives  $\widetilde{H^-}$ , and  $G_t := \widetilde{H^-}_t \circ \tilde{H}_t$  is a lift of the constant homotopy  $\pi^\circ$  starting at the identity; by uniqueness of lifts  $G_t = \text{id}$ , and symmetrically. So each  $\tilde{H}_t$  is a homeomorphism of  $S_2 \setminus W$ . Similarly  $G'_t := \iota^{-1} \tilde{H}_t \iota$  lifts  $F$  and starts at the identity, whence  $G'_t = \tilde{H}_t$ , that is,  $\tilde{H}_t$  commutes with  $\iota$ .

Extend by  $\tilde{H}(w, t) := w$  for  $w \in W$ . For continuity at  $(w, t_0)$ , let  $b := \pi(w)$  and let  $U' \ni b$  be an open disc with  $U' \cap B = \{b\}$ ; the sets  $U := \pi^{-1}(U')$  form a neighbourhood basis of  $w$ . Continuity of  $H$  at  $(b, t_0)$ , together with  $H(b, t_0) = b$ , gives an open  $V' \ni b$  and an interval  $J \ni t_0$  with  $H(V' \times J) \subset U'$ ; put  $V := \pi^{-1}(V')$ . For  $(x, t) \in V \times J$ : if  $x \in W$  then  $\tilde{H}(x, t) = x$  and  $\pi(x) = H(\pi x, t) \in U'$ , so  $x \in U$ ; if  $x \notin W$  then  $\pi \tilde{H}(x, t) = H(\pi x, t) \in U'$ , so  $\tilde{H}(x, t) \in U$ . Thus  $\tilde{H}(V \times J) \subset U$ .

Finally each  $\tilde{H}_t$  is a continuous bijection of the compact Hausdorff space  $S_2$ , hence a homeomorphism; the identities  $\pi \tilde{H}_t = H_t \pi$  and  $\iota \tilde{H}_t = \tilde{H}_t \iota$  hold on the dense set  $S_2 \setminus W$ , hence everywhere. The last display follows by applying the first identity to  $\tilde{H}_1$  and to  $\tilde{H}_1^{-1}$ .  $\square$

## 4.2. Preimages.

**Lemma 4.2.** *Let  $\alpha$  be an arc and  $c$  a simple closed curve in  $S^2 \setminus B$ .*

- (1) *If  $\alpha$  has two distinct endpoints then  $\pi^{-1}(\alpha)$  is a non-separating simple closed curve in  $S_2$ ; in particular it is essential.*
- (2) *If  $c$  is a 3|3 circle then  $\pi^{-1}(c)$  is a separating essential simple closed curve which splits  $S_2$  into two one-holed tori.*
- (3) *If  $\alpha$  is a loop based at  $b$  then  $\pi^{-1}(\alpha)$  is a wedge of two circles joined at  $w = \pi^{-1}(b)$ ; it is not a simple closed curve.*

*Proof.* (i) Over the interior of  $\alpha$  the map  $\pi$  is a double covering and over each endpoint it is one-to-one, so  $\pi^{-1}(\alpha)$  is obtained from two copies of the interior by gluing their ends in pairs at the two Weierstrass points: a circle, embedded because  $\alpha$  is. Its complement is  $\pi^{-1}(D)$ , where  $D = S^2 \setminus \alpha$  is an open disc containing the remaining four points of  $B$ ; the group  $\pi_1(D \setminus B)$  is generated by four meridians, each sent by  $\theta$  to the non-trivial element, so the cover is connected. A curve with connected complement does not separate.

(ii) The class  $[c]$  in  $H_1(S^2 \setminus B; \mathbb{Z}/2)$  is the sum of the three meridians on one side, so  $\theta([c]) \neq 0$  and  $\pi^{-1}(c)$  is connected: a simple closed curve double covering  $c$ . Write  $D_1, D_2$  for the two closed discs bounded by  $c$ , each containing three points of  $B$ . Each  $\pi^{-1}(D_i)$  is connected, again because a meridian is not killed by  $\theta$ , and Riemann–Hurwitz gives  $\chi(\pi^{-1}(D_i)) = 2 \cdot 1 - 3 = -1$ . A connected compact surface with one boundary circle has  $\chi = 1 - 2g$ , so  $g = 1$ : each  $\pi^{-1}(D_i)$  is a one-holed torus. Hence  $\pi^{-1}(c)$  separates and is essential.

(iii) Removing  $b$  from  $\alpha$  leaves an open arc whose preimage is two open arcs, and both ends of each of them converge to  $w$ ; so  $\pi^{-1}(\alpha)$  is the union of two circles meeting exactly at  $w$ .  $\square$

*Remark 4.3.* Part (iii) is the reason the cone metric cannot be used everywhere. The whole apparatus attached to  $\rho$  — unique closed geodesics, minimal position — speaks only about simple closed curves upstairs, and the preimage of a loop arc is not one. Families of arcs containing loops are handled exclusively by the cusped metric, through Theorem 3.13.

## 4.3. Projections.

**Lemma 4.4.** *Let  $a \subset S_2$  be an  $\iota$ -invariant essential simple closed curve, not assumed geodesic. Then exactly one of the following holds.*

- (1)  *$a \cap W$  consists of two points,  $a$  is non-separating,  $\pi(a)$  is an essential arc joining two distinct points of  $B$ , and  $\pi^{-1}(\pi(a)) = a$ .*
- (2)  *$a \cap W = \emptyset$ ,  $a$  is separating,  $\pi(a)$  is a 3|3 circle, and  $\pi^{-1}(\pi(a)) = a$ .*

*Proof.*  $\iota$ -invariance gives  $\pi^{-1}(\pi(a)) = a \cup \iota(a) = a$  at once. The restriction  $\iota|_a$  is an involution of a circle, not the identity (else  $a \subseteq W$ , a finite set). A non-trivial involution of a circle is either

orientation-reversing with exactly two fixed points, or orientation-preserving and free. Its fixed point set is  $a \cap W$ .

Case (i):  $a \cap W = \{w_1, w_2\}$ . Then  $b_i := \pi(w_i)$  are two distinct points of  $B$ , the quotient  $a/\iota$  is an arc, and  $\pi|_a$  factors through it injectively, so  $\pi(a)$  is an embedded arc with endpoints  $b_1, b_2$ ; its interior misses  $B$  because  $a \cap W$  consists of the two endpoints. It is essential by Observation 2.4. Non-separation follows as in Lemma 4.2(i).

Case (ii):  $a \cap W = \emptyset$ . Then  $\pi(a)$  is an embedded circle in  $S^2 \setminus B$ , bounding closed discs  $D_1, D_2$  with  $k_i := |D_i \cap B|$  and  $k_1 + k_2 = 6$ . If some  $k_i = 0$  then  $\pi^{-1}(D_i)$  is a disjoint union of two discs and  $a$  bounds a disc, contradicting essentiality. Since  $a$  is connected,  $\pi(a)$  does not lift, that is,  $\theta([\pi(a)]) \neq 0$ ; as  $[\pi(a)]$  is the sum of the  $k_i$  meridians inside  $D_i$ , each  $k_i$  is odd. If  $k_i = 1$  then  $\pi^{-1}(D_i)$  is connected with  $\chi = 2 \cdot 1 - 1 = 1$  and one boundary circle, hence a disc, again contradicting essentiality. Therefore  $k_1 = k_2 = 3$ . The two sets  $\pi^{-1}(\text{int } D_i)$  are connected and disjoint, so  $a$  separates.  $\square$

**4.4. Descent.** The next lemma is the heart of the matter: an isotopy *upstairs*, which is not assumed equivariant in any way, forces an isotopy *downstairs* which fixes the six marked points. The scheme of the proof is Winarski's combinatorial-topology proof of the Birman–Hilden theorem, in the exposition of Margalit and Winarski [1, the section “Winarski's proof: combinatorial topology”]; we give the argument in full because the step asserting that an innermost bigon injects under the covering requires, in our situation, the case analysis of Lemma 4.5 below, and because the annulus case requires the supplement given in the proof of Lemma 4.6.

Throughout this subsection  $c_1, c_2$  denote multicurves in  $S^2 \setminus B$ , by which we mean finite unions of disjoint simple closed curves, and  $\tilde{c}_i := \pi^{-1}(c_i)$ . Each  $\tilde{c}_i$  is an  $\iota$ -invariant multicurve disjoint from  $W$ , and  $\iota$  acts freely on it.

**Lemma 4.5** (Innermost bigons inject). *Let  $c_1 \neq c_2$  be transverse multicurves in  $S^2 \setminus B$  with  $c_1 \cap c_2 \neq \emptyset$ , and suppose  $\tilde{c}_1$  and  $\tilde{c}_2$  bound a bigon. Let  $\tilde{B}$  be an innermost bigon between  $\tilde{c}_1$  and  $\tilde{c}_2$ , with sides  $a \subset \tilde{c}_1$ ,  $b \subset \tilde{c}_2$  and corners  $x, y$ . Then*

$$\tilde{B} \cap \iota(\tilde{B}) = \emptyset.$$

*Consequently  $\tilde{B} \cap W = \emptyset$ ,  $\pi|_{\tilde{B}}$  is injective, and  $\pi(\tilde{B})$  is an embedded bigon between  $c_1$  and  $c_2$  whose interior and boundary are disjoint from  $B$ .*

*Proof.* Being innermost means  $\text{int } \tilde{B} \cap (\tilde{c}_1 \cup \tilde{c}_2) = \emptyset$ . We first record:

$$\text{int}(a) \cap \tilde{c}_2 = \emptyset, \quad \text{int}(b) \cap \tilde{c}_1 = \emptyset. \quad (3)$$

Indeed, if  $z \in \text{int}(a) \cap \tilde{c}_2$  then  $\tilde{c}_2$  crosses  $\tilde{c}_1$  transversally at  $z$  and one of the two local branches enters  $\text{int } \tilde{B}$ , contradicting innermostness. In particular  $\partial \tilde{B} \cap \tilde{c}_1 \cap \tilde{c}_2 = \{x, y\}$ .

*Step 1:*  $\iota(\tilde{B}) \neq \tilde{B}$ . If  $\iota(\tilde{B}) = \tilde{B}$  then  $\iota(\partial \tilde{B}) = \partial \tilde{B}$ , and since  $\iota(a) \subset \tilde{c}_1$  we get  $\iota(a) \subseteq \tilde{c}_1 \cap \partial \tilde{B} = a$ , so  $\iota(a) = a$ . But  $\iota$  acts freely on  $\tilde{c}_1$ , whereas every involution of an arc has a fixed point.

*Step 2:* *two distinct innermost bigons have disjoint interiors.* Let  $\tilde{B}'$  be another innermost bigon. Since  $\partial \tilde{B}' \subset \tilde{c}_1 \cup \tilde{c}_2$  is disjoint from  $\text{int } \tilde{B}$ , the set  $\tilde{B}' \cap \text{int } \tilde{B} = \text{int } \tilde{B}' \cap \text{int } \tilde{B}$  is open and closed in the connected set  $\text{int } \tilde{B}$ , hence empty or all of it; in the latter case the symmetric argument gives  $\text{int } \tilde{B} = \text{int } \tilde{B}'$  and  $\tilde{B} = \tilde{B}'$ . Applying this to  $\tilde{B}' = \iota(\tilde{B})$ , which is an innermost bigon because  $\iota$  preserves  $\tilde{c}_1$  and  $\tilde{c}_2$ , and using Step 1, we get  $\text{int } \tilde{B} \cap \iota(\tilde{B}) = \emptyset$  and, symmetrically,  $\tilde{B} \cap \iota(\text{int } \tilde{B}) = \emptyset$ . Therefore

$$\tilde{B} \cap \iota(\tilde{B}) \subseteq \partial \tilde{B} \cap \iota(\partial \tilde{B}). \quad (4)$$

*Step 3:*  $\partial \tilde{B} \cap \iota(\partial \tilde{B}) = \emptyset$ . Suppose  $z \in \partial \tilde{B}$  and  $\iota(z) \in \partial \tilde{B}$ ; note  $z \neq \iota(z)$  since  $\partial \tilde{B} \cap W = \emptyset$ . If  $z$  is a corner, then  $z \in \tilde{c}_1 \cap \tilde{c}_2$ , hence  $\iota(z) \in \tilde{c}_1 \cap \tilde{c}_2 \cap \partial \tilde{B} = \{x, y\}$  by (3), so  $\iota$  interchanges  $x$

and  $y$ . If  $z \in \text{int}(a)$ , then  $\iota(z) \in \tilde{c}_1 \cap \partial \tilde{B} = a$ , and  $\iota(z)$  is not a corner (otherwise  $z = \iota(\iota(z)) \in \tilde{c}_2$ , contradicting (3)), so  $\iota(z) \in \text{int}(a)$ ; the case  $z \in \text{int}(b)$  is symmetric. So one of the following occurs.

(a)  $\text{int}(a) \cap \iota(\text{int}(a)) \neq \emptyset$  (or the same for  $b$ ). Let  $C$  be the component of  $\tilde{c}_1$  containing  $a$ . As distinct components are disjoint,  $\iota(C) = C$ , and  $\iota|_C$  is a free involution of a circle. Identify  $C$  with  $\mathbb{R}/2\mathbb{Z}$  so that  $\iota(t) = t + 1$ , and write  $a = [s_1, s_2]$  with  $s_2 - s_1 = L \in (0, 2)$ . Then  $\text{int}(a) \cap \iota(\text{int}(a)) \neq \emptyset$  forces  $L > 1$ , whence  $s_1 + 1 \in \text{int}(a)$ . But  $s_1 = x \in \tilde{c}_2$ , so  $s_1 + 1 = \iota(x) \in \tilde{c}_2$ , contradicting (3).

(b)  $\iota(x) = y$ . Let  $C \subseteq \tilde{c}_1$  and  $D \subseteq \tilde{c}_2$  be the components containing  $a$  and  $b$ ; as  $x, y$  lie on both,  $\iota(C) = C$  and  $\iota(D) = D$ , and  $\iota$  acts freely on each. Now  $\iota(a)$  is an arc of  $C$  from  $y$  to  $x$ , so it is either  $a$  or the complementary arc  $C \setminus \text{int}(a)$ . It cannot be  $a$ : an involution of an arc interchanging its endpoints has an interior fixed point, while  $\iota$  is free on  $C$ . Hence  $\iota(a) = C \setminus \text{int}(a)$ , and by (3),  $\text{int}(\iota(a)) \cap \tilde{c}_2 = \iota(\text{int}(a) \cap \tilde{c}_2) = \emptyset$ . Therefore  $C \cap \tilde{c}_2 = \{x, y\}$ , and symmetrically  $D \cap \tilde{c}_1 = \{x, y\}$ . Since  $\iota(C) = C$  and  $\iota(D) = D$  with  $\iota$  free,  $\pi(C)$  and  $\pi(D)$  are embedded circles in  $S^2$  and

$$\pi(C) \cap \pi(D) = \pi(C \cap D) = \pi(\{x, y\}) = \{\pi(x)\}$$

is a single point, at which the two circles cross transversally,  $\pi$  being a local homeomorphism there. This is impossible: two transverse simple closed curves in  $S^2$  meet in an even number of points, because  $H_1(S^2; \mathbb{Z}/2) = 0$  makes their mod 2 intersection number vanish.

*Step 4: conclusion.* By (4) and Step 3,  $\tilde{B} \cap \iota(\tilde{B}) = \emptyset$ . A point of  $\tilde{B} \cap W$  would lie in  $\tilde{B} \cap \iota(\tilde{B})$ , so  $\tilde{B} \cap W = \emptyset$  and  $\pi$  is a local homeomorphism on a neighbourhood of  $\tilde{B}$ ; and  $\pi(z) = \pi(z')$  for  $z \neq z'$  in  $\tilde{B}$  would mean  $z' = \iota(z) \in \tilde{B} \cap \iota(\tilde{B})$ . Hence  $\pi|_{\tilde{B}}$  is injective, so  $\pi(\tilde{B})$  is an embedded closed disc bounded by the arcs  $\pi(a) \subset c_1$  and  $\pi(b) \subset c_2$ , which meet exactly at the two points  $\pi(x), \pi(y)$ . Finally a point of  $\pi(\tilde{B}) \cap B$  would have its unique preimage in  $\tilde{B} \cap W$ , which is empty.  $\square$

**Lemma 4.6** (Descent). *Let  $c_1, c_2$  be multicurves in  $S^2 \setminus B$  with no two components isotopic in  $\Sigma$  within either one of them, and suppose that  $\tilde{c}_1$  and  $\tilde{c}_2$  are isotopic in  $S_2$ . Then  $c_1$  and  $c_2$  are isotopic in  $\Sigma$ .*

*Proof.* We may assume  $c_1$  and  $c_2$  transverse, and induct on  $|c_1 \cap c_2|$ .

*Inductive step.* Suppose  $c_1 \cap c_2 \neq \emptyset$ . Then  $\tilde{c}_1 \cap \tilde{c}_2 \neq \emptyset$ . Isotopic multicurves have vanishing geometric intersection number, so  $\tilde{c}_1$  and  $\tilde{c}_2$  are not in minimal position, and by the bigon criterion for multicurves — the unnumbered statement in Farb–Margalit [2, §1.2.4], a variant of [2, Proposition 1.7] — two of their components bound a bigon. Among all bigons choose one,  $\tilde{B}$ , whose interior meets  $\tilde{c}_1 \cup \tilde{c}_2$  in as few points as possible. If that number were positive, an arc of  $\tilde{c}_1 \cup \tilde{c}_2$  would cross  $\text{int} \tilde{B}$ ; such an arc has both its endpoints on one side of  $\partial \tilde{B}$ , since the two multicurves are embedded, and an innermost such arc cuts off a bigon whose interior meets the multicurves in strictly fewer points. So  $\tilde{B}$  is innermost.

By Lemma 4.5,  $\pi(\tilde{B})$  is an embedded bigon between  $c_1$  and  $c_2$  disjoint from  $B$ . A closed neighbourhood of  $\pi(\tilde{B})$  disjoint from  $B$  exists by compactness, and the isotopy of  $c_1$  across  $\pi(\tilde{B})$  supported in it is an isotopy of  $S^2$  fixing  $B$  pointwise. It lowers  $|c_1 \cap c_2|$  by two and, by Lemma 4.1, replaces  $\tilde{c}_1$  by an isotopic multicurve. Induction applies.

*Base case.* Suppose  $c_1 \cap c_2 = \emptyset$ , so that  $\tilde{c}_1$  and  $\tilde{c}_2$  are disjoint and isotopic. We treat separately the two cases which occur in Theorem 4.10.

*Base case,  $\tilde{c}_1$  and  $\tilde{c}_2$  connected.* Two disjoint isotopic essential simple closed curves cobound an annulus, and here they cobound exactly one: cutting  $S_2$  along two disjoint isotopic *separating* curves — which is what the connected case amounts to, by Lemma 4.4(ii), since  $\tilde{c}_i$  is  $\iota$ -invariant,



essential and disjoint from  $W$  — leaves an annulus and two one-holed tori. Let  $A$  be that annulus. Then  $\iota(A) = A$ , since  $A$  is the unique annulus complementary component and  $\iota$  preserves  $\tilde{c}_1 \cup \tilde{c}_2$ .

*Base case,  $\tilde{c}_1$  and  $\tilde{c}_2$  each with two components.* This is the situation of the second case of Theorem 4.10, where  $\tilde{c}_i = \pi^{-1}(c_i)$  for a circle  $c_i$  separating  $B$  as  $2|4$ ; the preimage of such a circle has two components, exchanged by  $\iota$ , because the class of  $c_i$  is killed by  $\theta$ . Label the components  $1, 1'$  of  $\tilde{c}_1$  and  $2, 2'$  of  $\tilde{c}_2$ , so that  $\iota$  acts on the four curves by  $(1\ 1')(2\ 2')$ .

The four curves are disjoint, pairwise isotopic and non-separating, so they cut  $S_2$  into three annuli and one genus one surface  $G$  with two boundary circles, arranged in a cycle: consecutive curves cobound an annulus, except for one consecutive pair which cobounds  $G$ . Since  $G$  is the only complementary piece which is not an annulus,  $\iota(G) = G$ , so  $\partial G$  is an  $\iota$ -invariant pair of curves. As  $\iota$  maps components of  $\tilde{c}_i$  to components of  $\tilde{c}_i$ , a pair consisting of one component of each is not  $\iota$ -invariant; hence  $\partial G = \{1, 1'\}$  or  $\partial G = \{2, 2'\}$ . Relabelling if necessary, assume the former. Then the cyclic order of the four curves is

$$1, 2, 2', 1',$$

with  $G$  between  $1'$  and  $1$ , and the three annuli are

$$A(1, 2), \quad A(2, 2'), \quad A(2', 1').$$

Not every complementary annulus has one boundary circle in  $\tilde{c}_1$  and the other in  $\tilde{c}_2$  (see Remark 4.7 below); we use only the two outer ones.

Take then the two annuli

$$A := A(1, 2), \quad A' := A(2', 1').$$

Each has one boundary circle in  $\tilde{c}_1$  and the other in  $\tilde{c}_2$ , and  $\iota(A) = A'$ , because  $\partial A = \{1, 2\}$  is carried to  $\{1', 2'\} = \partial A'$  and  $A \neq A'$ . In particular  $\iota$  has no fixed point in  $A \cup A'$ , so  $\pi|_A$  is injective,  $\pi(A) = \pi(A')$  is an annulus in  $S^2 \setminus B$  cobounded by  $c_1$  and  $c_2$ , and pushing  $c_1$  across it is an isotopy of  $S^2$  fixing  $B$  pointwise which carries  $c_1$  to  $c_2$ . This settles the second case, and the remainder of the proof concerns the first.

So let  $A_1, \dots, A_n$  be the complementary annuli in the connected case, each having one boundary circle in  $\tilde{c}_1$  and the other in  $\tilde{c}_2$ . We claim each  $\pi(A_m)$  is an annulus containing no point of  $B$ .

The involution  $\iota$  permutes the  $A_m$ . If  $\iota(A_m) = A_{m'} \neq A_m$  then  $A_m$  contains no fixed point of  $\iota$ , so  $\pi|_{A_m}$  is injective and  $\pi(A_m) \cong A_{m'}$  is an annulus disjoint from  $B$ . Suppose then  $\iota(A_m) = A_m$ . Since one boundary circle lies in  $\tilde{c}_1$  and the other in  $\tilde{c}_2$ , and  $\iota$  preserves each  $\tilde{c}_i$ , the involution  $\iota$  cannot interchange the two boundary circles; being orientation preserving on  $A_m$  it therefore preserves the orientation of each of them, and an orientation-preserving involution of a circle with a fixed point is the identity, which is impossible here. So  $\iota$  acts freely on  $\partial A_m$ , each boundary circle has a circle as quotient, and the two quotients are distinct. Hence the base surface of  $A_m/\iota$  is a connected compact surface with exactly two boundary circles, so  $\chi \leq 0$ . On the other hand, if  $\iota$  has  $k$  fixed points in  $\text{int } A_m$ , multiplicativity of the orbifold Euler characteristic gives

$$\chi(\text{base}) - k(1 - \tfrac{1}{2}) = \chi^{\text{orb}}(A_m/\iota) = \tfrac{1}{2}\chi(A_m) = 0,$$

so  $\chi(\text{base}) = k/2 \geq 0$ . Therefore  $k = 0$ : the action is free and  $\pi(A_m) = A_m/\iota$  is an annulus containing no branch point.

The images  $\pi(A_m)$  are annuli in  $S^2 \setminus B$  cobounded by the components of  $c_1$  and  $c_2$ , so pushing  $c_1$  across them is an isotopy of  $S^2$  fixing  $B$  pointwise carrying  $c_1$  to  $c_2$ .  $\square$

*Remark 4.7.* In the second base case above it is *not* true that every complementary annulus has one boundary circle in  $\tilde{c}_1$  and the other in  $\tilde{c}_2$ : the middle annulus  $A(2, 2')$  is bounded by the two components of  $\tilde{c}_2$ . Two disjoint isotopic multicurves need not cobound annuli pairing their

components across, as that nested configuration shows, which is why the proof uses only the two outer annuli. This is worth stating because the naive form of the assertion is tempting and false.

#### 4.5. From circles back to arcs.

**Lemma 4.8.** *Let  $D$  be a closed disc and  $p \neq q$  two interior points. Any two embedded arcs in  $D$  with endpoints  $p, q$  and interiors contained in  $\text{int } D \setminus \{p, q\}$  are carried onto one another by an ambient isotopy of  $D$  which is the identity on  $\partial D$  and fixes  $p$  and  $q$ .*

*Proof.* Work in  $\text{int } D \cong \mathbb{R}^2$ . We use the following fact: any two embedded arcs in  $\mathbb{R}^2$  with the same pair of endpoints are carried onto one another by a compactly supported ambient isotopy. This is the arc case, relative to the endpoints, of Epstein's isotopy theorem [10]; alternatively it follows from the change of coordinates principle of Farb–Margalit [2, §1.3]. Such an isotopy carries the endpoint set  $\{p, q\}$  to itself, and since it starts at the identity and  $\{p, q\}$  is discrete it fixes  $p$  and  $q$  throughout. Being compactly supported in  $\text{int } D$ , it is the identity near  $\partial D$  and extends by the identity.  $\square$

**Lemma 4.9.** *Let  $\alpha$  be an arc with distinct endpoints  $p, q$ , let  $N(\alpha)$  be a closed regular neighbourhood of  $\alpha$  in  $S^2$  chosen so that  $N(\alpha) \cap B = \{p, q\}$ , and put  $c_\alpha := \partial N(\alpha)$ , a  $2|4$  circle.*

- (1)  $\pi^{-1}(N(\alpha))$  is an annulus with core  $\pi^{-1}(\alpha)$ , and  $\pi^{-1}(c_\alpha)$  is the union of its two boundary circles, each isotopic in  $S_2$  to  $\pi^{-1}(\alpha)$ .
- (2) If  $c_\alpha$  and  $c_\beta$  are isotopic in  $\Sigma$ , for arcs  $\alpha, \beta$  with distinct endpoints, then  $\alpha$  and  $\beta$  are isotopic in  $\Sigma$ .

*Proof.* (i)  $N(\alpha)$  is a disc containing exactly two branch points, so  $\chi(\pi^{-1}(N(\alpha))) = 2 \cdot 1 - 2 = 0$ ; the preimage is connected because a meridian is not killed by  $\theta$ ; and  $c_\alpha$  encircles an even number of branch points, so  $\theta([c_\alpha]) = 0$  and  $\pi^{-1}(c_\alpha)$  has two components. A connected compact orientable surface with  $\chi = 0$  and two boundary circles is an annulus. Its core is  $\pi^{-1}(\alpha)$ , which is a simple closed curve by Lemma 4.2(i), and each boundary circle of an annulus is isotopic to its core.

(ii) Let  $h_1$  be a homeomorphism of  $S^2$  isotopic to the identity through homeomorphisms fixing  $B$  pointwise, with  $h_1(c_\alpha) = c_\beta$ . Then  $h_1(N(\alpha))$  is one of the two closed discs bounded by  $c_\beta$  and it contains exactly the two branch points  $h_1(p) = p$  and  $h_1(q) = q$ ; the other disc contains four. Hence  $h_1(N(\alpha)) = N(\beta)$  and  $\{p, q\} = \{p', q'\}$ , the endpoints of  $\beta$ . Now  $h_1(\alpha)$  and  $\beta$  are two embedded arcs in the disc  $N(\beta)$  with endpoints  $p, q$  and interiors in the interior, so Lemma 4.8 produces an ambient isotopy of  $N(\beta)$ , identity on the boundary and fixing  $p, q$ , carrying  $h_1(\alpha)$  to  $\beta$ . Extending by the identity and composing with  $\{h_t\}$  gives  $\alpha \simeq \beta$  in  $\Sigma$ .  $\square$

#### 4.6. The bijection.

**Theorem 4.10.** *The assignment  $\Psi([\alpha]) := [\pi^{-1}(\alpha)]$  is a well-defined bijection*

$$\Psi: \mathcal{A}^\# \sqcup \mathcal{C} \longrightarrow \mathcal{S},$$

*carrying  $\mathcal{A}^\#$  onto the set of non-separating classes and  $\mathcal{C}$  onto the set of separating classes. Its inverse sends a class  $[a]$  to  $[\pi(a_\iota)]$ , where  $a_\iota$  is any  $\iota$ -invariant representative.*

*Proof.* Well defined. By Lemma 4.2,  $\pi^{-1}$  of a representative is an essential simple closed curve, non-separating in the arc case and separating in the  $3|3$  case. If  $\alpha \simeq \alpha'$  in  $\Sigma$  then Lemma 4.1 produces an ambient isotopy of  $S_2$  carrying  $\pi^{-1}(\alpha)$  to  $\pi^{-1}(\alpha')$ .

*Surjective.* Let  $[a] \in \mathcal{S}$ . Give  $S_2$  the metric  $\rho$  of Proposition 3.4. By (G1) of Fact 3.5 the free homotopy class of  $a$  contains a unique closed geodesic  $\gamma$ , which is simple; by (G3)  $\gamma$  is isotopic to  $a$ . Since  $\rho$  comes from a complex structure for which  $\iota$  is the hyperelliptic involution, Fact 3.6

gives  $\iota(\gamma) = \gamma$ . Lemma 4.4 then says  $\pi(\gamma)$  is an essential arc with distinct endpoints or a 3|3 circle, with  $\pi^{-1}(\pi(\gamma)) = \gamma$ . Hence  $\Psi([\pi(\gamma)]) = [\gamma] = [a]$ .

*Injective.* Suppose  $\Psi([\alpha]) = \Psi([\alpha'])$ . If both classes lie in  $\mathcal{C}$ , then  $\alpha, \alpha'$  are 3|3 circles whose preimages are isotopic in  $S_2$ , and Lemma 4.6 applied to the one-component multicurves gives  $\alpha \simeq \alpha'$ . If both lie in  $\mathcal{A}^\neq$ , pass to the 2|4 circles: by Lemma 4.9(i) the multicurves  $\pi^{-1}(c_\alpha)$  and  $\pi^{-1}(c_{\alpha'})$  are the boundaries of annulus neighbourhoods of  $\pi^{-1}(\alpha)$  and  $\pi^{-1}(\alpha')$ ; an ambient isotopy of  $S_2$  carrying the first core to the second carries a regular neighbourhood of the first to a regular neighbourhood of the second, and regular neighbourhoods of a simple closed curve in a surface are unique up to ambient isotopy, by the tubular neighbourhood theorem [4], so  $\pi^{-1}(c_\alpha)$  and  $\pi^{-1}(c_{\alpha'})$  are isotopic. The two components of  $\pi^{-1}(c_\alpha)$  are not isotopic to each other in  $\Sigma$  — their images  $c_\alpha$  has one component — so Lemma 4.6 applies and gives  $c_\alpha \simeq c_{\alpha'}$ , whence  $\alpha \simeq \alpha'$  by Lemma 4.9(ii). Finally the two types cannot mix: an equality  $\Psi([\alpha]) = \Psi([c])$  with  $\alpha \in \mathcal{A}^\neq$ ,  $c \in \mathcal{C}$  would equate a non-separating with a separating class.

The description of the inverse now follows: if  $a_\iota$  is an  $\iota$ -invariant representative of  $[a]$ , then  $\pi^{-1}(\pi(a_\iota)) = a_\iota$  by Lemma 4.4, so  $\Psi([\pi(a_\iota)]) = [a]$ .  $\square$

Theorem 1.4 is exactly Theorem 4.10.

## 5. THE INTERSECTION NUMBER FORMULA

Throughout this section  $a \neq b$  are isotopy classes of *non-loop* essential arcs in  $\Sigma$ , and

$$A := [\pi^{-1}(a)], \quad B := [\pi^{-1}(b)]$$

are the corresponding classes of non-separating simple closed curves in  $S_2$  furnished by Lemma 4.2(i). We write  $i(A, B)$  for their geometric intersection number. The restriction to non-loop arcs is essential and not a convenience: by Lemma 4.2(iii) the preimage of a loop arc is a wedge of two circles, so neither side of the formula below is defined for it.

### 5.1. The local count.

**Lemma 5.1** (Local count). *Let  $\alpha \in a$ ,  $\beta \in b$  be transverse, and suppose that  $\tilde{\alpha} := \pi^{-1}(\alpha)$  and  $\tilde{\beta} := \pi^{-1}(\beta)$  are transverse as well. Then*

$$|\tilde{\alpha} \cap \tilde{\beta}| = 2j(\alpha, \beta) + e(a, b).$$

*Proof.* The set  $\alpha \cap \beta$  is the disjoint union of  $\text{int } \alpha \cap \text{int } \beta$  and the set of shared endpoints, and  $\tilde{\alpha} \cap \tilde{\beta} = \pi^{-1}(\alpha \cap \beta)$  because  $\tilde{\alpha} = \pi^{-1}(\alpha)$  and  $\tilde{\beta} = \pi^{-1}(\beta)$  are full preimages. Over a point of  $\text{int } \alpha \cap \text{int } \beta$ , which lies in  $S^2 \setminus B$ , the map  $\pi$  is a two-sheeted covering, so there are exactly two preimages; there are  $j(\alpha, \beta)$  such points. A shared endpoint is a point of  $B$  and has exactly one preimage, a Weierstrass point; there are  $e(a, b)$  such points, by Lemma 2.10. Summing gives the stated count.  $\square$

Two remarks on the hypotheses. First, the identity is a count for the two *given* representatives; it does not by itself assert that either side is minimal in its class. Second, the hypothesis that the preimages be transverse cannot be dropped, and the only place it can fail is over a shared endpoint:

**Lemma 5.2** (The lift of an arc at a Weierstrass point). *Let  $\alpha$  be an arc with an endpoint at  $b \in B$ , let  $w = \pi^{-1}(b)$ , and let  $u$  be a coordinate at  $b$  in which  $\pi$  has the local model  $z \mapsto z^2$ . Write  $\alpha$  near  $b$  as  $u = u_\alpha(t)$ ,  $t \in [0, \varepsilon)$ , with  $u_\alpha(0) = 0$  and  $u'_\alpha(0) \neq 0$ , and set  $\theta_\alpha := \arg u'_\alpha(0)$ .*

- (1) *Near  $w$  the set  $\pi^{-1}(\alpha)$  is a smooth embedded arc through  $w$  whose tangent line at  $w$  has direction  $\theta_\alpha/2 \bmod \pi$ .*

- (2) If  $\beta$  is a second arc with an endpoint at  $b$ , then  $\pi^{-1}(\alpha)$  and  $\pi^{-1}(\beta)$  are transverse at  $w$  if and only if  $\theta_\alpha \not\equiv \theta_\beta \pmod{2\pi}$ .

*Proof.* (i) Write  $u_\alpha(t) = tc(t)$  with  $c$  smooth and  $c(0) = u'_\alpha(0) \neq 0$ ; this is Hadamard's factorisation of a smooth function vanishing at 0. A lift must satisfy  $z(s)^2 = u_\alpha(s^2)$  for a parameter  $s$ , and the naive choice  $z = \sqrt{u_\alpha(t)}$  is *not* differentiable at  $t = 0$ : it behaves like  $\sqrt{t}$ . Substituting  $t = s^2$  removes the branch point. Choose a smooth branch  $\sqrt{c}$  of the square root of  $c$  near 0, which exists because  $c(0) \neq 0$ , and put

$$z(s) := s \sqrt{c(s^2)}, \quad s \in (-\varepsilon^{1/2}, \varepsilon^{1/2}).$$

Then  $z$  is smooth,  $z(s)^2 = s^2 c(s^2) = u_\alpha(s^2)$ , so  $z$  parametrises a lift;  $z(0) = 0$  and  $z'(0) = \sqrt{c(0)} \neq 0$ , so the lift is regular at  $w$  and its tangent direction is  $\arg \sqrt{c(0)} = \theta_\alpha/2 \pmod{\pi}$ . Finally  $z(-s) = -z(s)$ , so the two halves  $s > 0$  and  $s < 0$  are the two sheets of  $\pi^{-1}$  over  $\alpha$  and together form a single smooth arc through  $w$ .

(ii) By (i) the two tangent lines at  $w$  have directions  $\theta_\alpha/2$  and  $\theta_\beta/2$  modulo  $\pi$ , and these agree if and only if  $\theta_\alpha \equiv \theta_\beta \pmod{2\pi}$ .  $\square$

The halving in Lemma 5.2 has a consequence worth recording, since it is the opposite of what the downstairs picture suggests: if  $\alpha$  and  $\beta$  leave  $b$  in *opposite* directions,  $\theta_\beta = \theta_\alpha + \pi$ , then upstairs the two tangent lines differ by  $\pi/2$ , that is, the lifts meet *orthogonally*. The set of downstairs departure directions which are forbidden is therefore a single one,  $\theta_\beta = \theta_\alpha$ .

## 5.2. Normalising the departure directions.

**Lemma 5.3** (Normalisation). *Let  $\alpha \in a$ ,  $\beta \in b$  be transverse. Then there are transverse  $\alpha_1 \in a$ ,  $\beta_1 \in b$  such that  $\pi^{-1}(\alpha_1)$  and  $\pi^{-1}(\beta_1)$  are transverse and*

$$j(\alpha_1, \beta_1) = j(\alpha, \beta).$$

*Proof.* Over  $S^2 \setminus B$  the map  $\pi$  is a local diffeomorphism, so at every point of  $\text{int } \alpha \cap \text{int } \beta$  the preimages inherit distinct tangent lines from  $\alpha$  and  $\beta$ ; and  $\pi^{-1}(\alpha) \cap \pi^{-1}(\beta)$  is finite because  $\alpha \cap \beta$  is. By Lemma 5.2(ii) the preimages can fail to be transverse only at a Weierstrass point over a shared endpoint, and only when the two departure directions there agree. There are at most two shared endpoints, so it suffices to correct one of them and to repeat.

Let  $b$  be a shared endpoint with  $\theta_\alpha = \theta_\beta$ . Fix a coordinate  $u$  at  $b$  and choose  $r > 0$  so small that the closed disc  $\overline{D_r} = \{|u| \leq r\}$  contains no point of  $B$  other than  $b$  and no point of  $\alpha \cap \beta$  other than  $b$ ; this uses the finiteness clause of Convention 2.8 together with compactness. Shrinking  $r$  further, we may assume that  $\alpha$  and  $\beta$  meet  $\overline{D_r}$  in a single arc each and that on  $\overline{D_r}$  each of them is a *radial graph*: there are smooth functions  $f, g$  on  $[0, r]$  with

$$\alpha \cap \overline{D_r} = \{\varrho e^{if(\varrho)} : 0 \leq \varrho \leq r\}, \quad \beta \cap \overline{D_r} = \{\varrho e^{ig(\varrho)} : 0 \leq \varrho \leq r\},$$

which is possible because  $|u_\alpha(t)| = t|c(t)|$  is strictly increasing near 0, so the radius is a legitimate parameter.

Pick a smooth cut-off  $\chi : [0, \infty) \rightarrow [0, 1]$  with  $\chi \equiv 1$  on  $[0, r/3]$  and  $\chi \equiv 0$  on  $[2r/3, \infty)$ , and for  $\varphi \in \mathbb{R}$  let

$$\Phi_\varphi(u) := e^{i\varphi\chi(|u|)} u,$$

extended by the identity outside  $D_r$ . Each  $\Phi_\varphi$  is a diffeomorphism of  $S^2$ ; the family  $\Phi_{t\varphi}$ ,  $t \in [0, 1]$ , is an isotopy from the identity; it fixes  $b$ , and it fixes every other point of  $B$  because those lie outside  $D_r$ . Hence  $\Phi_\varphi$  is admissible for Convention 2.1, and  $\alpha_1 := \Phi_\varphi(\alpha)$  is isotopic to  $\alpha$  with  $\theta_{\alpha_1} = \theta_\alpha + \varphi$ .

It remains to choose  $\varphi$  so that no new intersection is created. Inside  $\overline{D_r}$  the curve  $\alpha_1$  is the radial graph of  $f + \varphi\chi$ , so a point of  $\alpha_1 \cap \beta$  in  $\overline{D_r}$  with radius  $\varrho > 0$  satisfies

$$h(\varrho) + \varphi\chi(\varrho) \equiv 0 \pmod{2\pi}, \quad h := f - g.$$

Since  $h(0) = 0$  and  $h$  is continuous,  $M := \max_{[0,r]} |h|$  tends to 0 as  $r \rightarrow 0$ ; shrinking  $r$  once more we may assume  $M < \pi$ . Then for every  $\varphi$  with  $0 < |\varphi| < 2\pi - 2M$  and  $\varphi$  of the appropriate sign the left-hand side is bounded away from  $2\pi\mathbb{Z}$  on  $(0, r]$ , so  $\alpha_1 \cap \beta \cap \overline{D_r} = \{b\}$ ; outside  $D_r$  nothing has moved. Consequently  $\alpha_1 \cap \beta = \alpha \cap \beta$  as sets, whence  $j(\alpha_1, \beta) = j(\alpha, \beta)$  and  $e$  is unchanged. Choosing in addition  $\varphi \neq 0$  we obtain  $\theta_{\alpha_1} \neq \theta_\beta$ , and by Lemma 5.2(ii) the preimages are transverse over  $b$ .  $\square$

*Remark 5.4.* Lemma 5.3 is a statement about a *pair* of arcs, and only pairs are needed below, because (1) is a statement about a pair. It should not be read as a normalisation for a family: rotating several arcs through a common endpoint by different angles cannot be achieved by one ambient isotopy, since the derivative at that point is a single linear map. Families are handled instead by Theorem 3.13, where no perturbation occurs at all: the geodesic representative of a class is a single well-defined curve, so a family of classes realises all its pairwise minima simultaneously by fiat.

### 5.3. The formula.

**Theorem 5.5.** *For distinct isotopy classes  $a, b$  of non-loop essential arcs in  $\Sigma$ ,*

$$i(A, B) = 2\hat{j}(a, b) + e(a, b). \quad (1)$$

*Moreover the projections to  $\Sigma$  of the  $\rho$ -geodesic representatives of  $A$  and  $B$  are representatives of  $a$  and  $b$  realising  $\hat{j}(a, b)$ .*

*Proof.* Write  $\hat{j} = \hat{j}(a, b)$  and  $e = e(a, b)$ .

*Upper bound.* Let  $(\alpha_0, \beta_0)$  be a transverse pair realising  $\hat{j}$ , which exists by Definition 2.9. Apply Lemma 5.3 to obtain a transverse pair  $(\alpha_1, \beta_1)$  with transverse preimages and  $j(\alpha_1, \beta_1) = \hat{j}$ . Then  $\pi^{-1}(\alpha_1)$  and  $\pi^{-1}(\beta_1)$  are transverse representatives of  $A$  and  $B$ , so

$$i(A, B) \leq |\pi^{-1}(\alpha_1) \cap \pi^{-1}(\beta_1)| = 2\hat{j} + e$$

by Lemma 5.1.

*Lower bound.* Here the representatives are not chosen downstairs at all. By Proposition 3.4 the pullback  $\rho = \pi^*\bar{\rho}$  is a smooth hyperbolic metric on the closed surface  $S_2$ , so by (G1) of Fact 3.5 the classes  $A$  and  $B$  have unique geodesic representatives  $\tilde{\alpha}^*$  and  $\tilde{\beta}^*$ , and these are simple. By Fact 3.6 every simple closed geodesic of  $(S_2, \rho)$  is  $\iota$ -invariant, so Lemma 4.4 applies to each of them; as  $A$  and  $B$  are non-separating by Lemma 4.2(i), case (i) of that lemma holds, and

$$\alpha^* := \pi(\tilde{\alpha}^*), \quad \beta^* := \pi(\tilde{\beta}^*)$$

are essential arcs joining two distinct points of  $B$ , with  $\pi^{-1}(\alpha^*) = \tilde{\alpha}^*$  and  $\pi^{-1}(\beta^*) = \tilde{\beta}^*$ . Consequently  $\Psi[\alpha^*] = [\tilde{\alpha}^*] = A = \Psi(a)$ , so  $[\alpha^*] = a$  by the injectivity of  $\Psi$  in Theorem 4.10, and likewise  $[\beta^*] = b$ .

Since  $A \neq B$  we have  $\tilde{\alpha}^* \neq \tilde{\beta}^*$ , so these are two distinct simple closed geodesics: they meet in finitely many points, they are nowhere tangent, and by (G2) of Fact 3.5 they are in minimal position. Hence they are transverse; projecting,  $\alpha^*$  and  $\beta^*$  are transverse as well, since  $\pi$  is a local diffeomorphism off  $W$  and  $|\alpha^* \cap \beta^*| \leq |\tilde{\alpha}^* \cap \tilde{\beta}^*|$  is finite, no condition being imposed at a shared endpoint. Lemma 5.1 now applies to the pair  $(\alpha^*, \beta^*)$ , whose preimages are  $\tilde{\alpha}^*, \tilde{\beta}^*$ , and minimal position gives

$$i(A, B) = |\tilde{\alpha}^* \cap \tilde{\beta}^*| = 2j(\alpha^*, \beta^*) + e \geq 2\hat{j} + e,$$

the last step because  $j(\alpha^*, \beta^*) \geq \hat{j}$  by the definition of  $\hat{j}$ .

The two bounds give (1), and comparing them forces  $j(\alpha^*, \beta^*) = \hat{j}$ .  $\square$

*Remark 5.6.* The two halves of the proof are asymmetric, and the asymmetry is not an artefact. In the lower bound no normalisation is needed: distinct closed geodesics of a smooth metric are automatically transverse, because a geodesic is determined by a point and a direction. In the upper bound one quantifies over arbitrary representatives, where no such rigidity is available, and this is exactly the work done by Lemma 5.3.

**Corollary 5.7** (Adjacency criterion). *With  $a, b$  as above,*

$$i(A, B) \leq 1 \iff \hat{j}(a, b) = 0 \text{ and } e(a, b) \leq 1.$$

*Proof.* Both  $\hat{j}$  and  $e$  are non-negative integers. If  $i(A, B) \leq 1$  then  $2\hat{j} \leq 1$  by (1), so  $\hat{j} = 0$  and  $e = i(A, B) \leq 1$ . Conversely  $\hat{j} = 0$  and  $e \leq 1$  give  $i(A, B) = e \leq 1$ .  $\square$

## 6. FROM $C_1(S_2)$ TO $X$

Write  $K := C_1(S_2)$  and let  $K' \subset K$  be the full subcomplex spanned by the non-separating vertices. This section identifies  $K'$  with  $X$  and shows that  $K'$  is a strong deformation retract of  $K$ .

### 6.1. The model.

**Proposition 6.1.**  *$K'$  and  $X$  are isomorphic as simplicial complexes.*

*Proof.* By Theorem 4.10 the map  $\Psi$  is a bijection from the set of isotopy classes of non-loop arcs in  $\Sigma$  onto the set of isotopy classes of non-separating essential simple closed curves in  $S_2$ ; that is, it is a bijection from the vertex set of  $X$  onto the vertex set of  $K'$ .

Let  $a \neq b$  be non-loop classes and  $A = \Psi(a)$ ,  $B = \Psi(b)$ . By Corollary 5.7,

$$i(A, B) \leq 1 \iff \hat{j}(a, b) = 0 \text{ and } e(a, b) \leq 1.$$

The right-hand side is precisely adjacency in  $G$ : by Theorem 3.13 the condition  $\hat{j}(a, b) = 0$  says that  $a$  and  $b$  can be realised with disjoint interiors, and by Lemma 2.10 the condition  $e(a, b) \leq 1$  says  $\text{ep}(a) \neq \text{ep}(b)$ , two distinct two-element subsets of  $B$  meeting in at most one point. The left-hand side is adjacency in  $K'$ . So  $\Psi$  is an isomorphism of graphs from  $G$  onto the 1-skeleton of  $K'$ .

Both complexes are flag:  $X$  by definition, and  $K$  by definition, whence also its full subcomplex  $K'$  — if the vertices of a finite set lie in  $K'$  and are pairwise adjacent in  $K'$ , they are pairwise adjacent in  $K$ , so they span a simplex of  $K$ , and that simplex lies in the full subcomplex  $K'$ . An isomorphism of the 1-skeleta of two flag complexes is an isomorphism of the complexes.  $\square$

**Proposition 6.2.**  $\dim X = 11$ , and the facets of  $X$  are exactly the isotopy classes of triangulations of  $S^2$  with vertex set  $B$  whose underlying graph is simple.

*Proof.* A simplex of  $X$  is a finite set of non-loop classes which are pairwise adjacent in  $G$ . By Theorem 3.13 the geodesic representatives of the classes in such a set have pairwise disjoint interiors, so their union is an embedded graph in  $S^2$  with vertex set contained in  $B$ ; adjacency in  $G$  forbids two members with the same endpoint pair, so the graph has no multiple edges, and non-loop forbids loops. Conversely the classes of the edges of an embedded simple graph with vertex set in  $B$  are pairwise adjacent in  $G$ . Simplices of  $X$  therefore correspond to isotopy classes of embedded simple graphs with vertex set contained in  $B$ , a  $k$ -simplex to a graph with  $k + 1$  edges.



For a simple graph with  $V \geq 3$  vertices embedded in  $S^2$  one has  $E \leq 3V - 6$ , so here  $E \leq 3 \cdot 6 - 6 = 12$  and  $\dim X \leq 11$ . The octahedral triangulation of  $S^2$  with vertex set  $B$  realises  $E = 12$ , so  $\dim X = 11$ . Equality  $E = 3V - 6$  holds if and only if every complementary face is a triangle, that is, if and only if the graph is a triangulation.  $\square$

*Remark 6.3.* A simplex of  $X$  is an isotopy class of *embedded* graph, not an abstract graph: two embeddings of the same abstract graph which are not isotopic rel  $B$  are different simplices. Accordingly  $X$  has infinitely many facets, although these fall into finitely many  $\text{Mod}(S^2, B)$ -orbits, and the number of combinatorial types of six-vertex triangulations of  $S^2$  is not the number of facets.

## 6.2. Deleting the separating vertices.

**Lemma 6.4.** *Distinct separating vertices of  $K$  are never adjacent.*

*Proof.* Let  $\sigma \neq \sigma'$  be separating. A separating curve is null-homologous, so the algebraic intersection number  $\hat{i}(\sigma, \sigma')$  vanishes, and  $i(\sigma, \sigma') \equiv \hat{i}(\sigma, \sigma') \equiv 0 \pmod{2}$ . Adjacency would force  $i(\sigma, \sigma') \leq 1$ , hence  $i(\sigma, \sigma') = 0$ .

So suppose  $i(\sigma, \sigma') = 0$ . Now  $\sigma$  splits  $S_2$  into two one-holed tori  $T_1, T_2$ , and  $\sigma'$ , being disjoint from  $\sigma$ , lies in some  $T_m$ . An essential simple closed curve in a one-holed torus is either isotopic to the boundary, in which case  $\sigma' = \sigma$ , excluded; or non-separating in  $T_m$ , in which case  $[\sigma'] \neq 0$  in  $H_1(T_m; \mathbb{Z})$ . The inclusion induced map  $H_1(T_m; \mathbb{Z}) \rightarrow H_1(S_2; \mathbb{Z})$  is the inclusion of a direct summand, hence injective, so  $[\sigma'] \neq 0$  in  $H_1(S_2; \mathbb{Z})$ , contradicting that  $\sigma'$  separates.  $\square$

**Corollary 6.5.** *Every simplex of  $K$  contains at most one separating vertex. Consequently, for separating vertices  $\sigma \neq \sigma'$ ,*

$$\text{lk}_K(\sigma) \subseteq K', \quad \overline{\text{St}}(\sigma) \cap K' = \text{lk}_K(\sigma), \quad \overline{\text{St}}(\sigma) \cap \overline{\text{St}}(\sigma') = \text{lk}_K(\sigma) \cap \text{lk}_K(\sigma') \subseteq K'.$$

*In particular each  $\text{lk}_K(\sigma)$  lies in  $K'$  irrespective of which other separating vertices are deleted.*

*Proof.* If a simplex contained two separating vertices they would be adjacent, contradicting Lemma 6.4. If  $\tau \in \text{lk}_K(\sigma)$  then  $\tau \cup \{\sigma\}$  is a simplex, so  $\tau$  contains no separating vertex and lies in the full subcomplex  $K'$ . The remaining two identities follow because a simplex of  $\overline{\text{St}}(\sigma)$  either contains  $\sigma$  or lies in  $\text{lk}_K(\sigma)$ .  $\square$

**Lemma 6.6.** *For every separating vertex  $\sigma$ , with  $T_1, T_2$  the two one-holed tori cut off by  $\sigma$ ,*

$$\text{lk}_K(\sigma) = C_1(T_1) * C_1(T_2),$$

*where  $C_1(T_m)$  is the Farey complex. In particular  $\text{lk}_K(\sigma)$  is contractible.*

*Proof.* For any essential simple closed curve  $c$  we have  $i(c, \sigma) \equiv \hat{i}(c, \sigma) = 0 \pmod{2}$ , so  $i(c, \sigma) \leq 1$  forces  $i(c, \sigma) = 0$ : the vertices of  $\text{lk}_K(\sigma)$  are the classes of essential curves disjoint from  $\sigma$  and not isotopic to it, that is, the essential non-boundary-parallel curves in  $T_1$  or in  $T_2$ . Two such curves lying on opposite sides are disjoint, hence adjacent, which gives the join decomposition.

In a one-holed torus the essential non-boundary-parallel isotopy classes correspond to slopes  $p/q \in \mathbb{Q} \cup \{\infty\}$  with  $i(p/q, r/s) = |ps - qr|$ , so the condition  $i \leq 1$  is  $|ps - qr| = 1$  and  $C_1(T_m)$  is the Farey complex, which is contractible: it is the simplicial model for  $\mathbb{H}^2$  obtained from the ideal triangulation by the Farey tessellation. A join with a contractible factor is contractible.  $\square$

**Proposition 6.7.**  *$K$  is the pushout, in simplicial complexes and hence in spaces, of*

$$K' \longleftarrow \coprod_{\sigma} \text{lk}_K(\sigma) \hookrightarrow \coprod_{\sigma} \overline{\text{St}}(\sigma),$$

*the coproducts being over all separating vertices.*

*Proof.* Every simplex of  $K$  either contains no separating vertex, and then lies in  $K'$ , or contains exactly one,  $\sigma$ , by Corollary 6.5, and then lies in  $\overline{\text{St}}(\sigma)$ . By the same corollary the intersection of two distinct stars, and the intersection of a star with  $K'$ , are exactly the links, which are the objects being glued.  $\square$

**Theorem 6.8.**  *$K'$  is a strong deformation retract of  $K$ . In particular  $C_1(S_2) \simeq X$ .*

*Proof.* Fix a separating vertex  $\sigma$ . By Lemma 6.6 the link  $\text{lk}_K(\sigma)$  is contractible, and  $\overline{\text{St}}(\sigma) = \sigma * \text{lk}_K(\sigma)$  is a cone, hence contractible; so the inclusion  $\text{lk}_K(\sigma) \hookrightarrow \overline{\text{St}}(\sigma)$  is a homotopy equivalence of CW complexes. A pair of CW complexes satisfies the homotopy extension property [3, Proposition 0.16], so by [3, Corollary 0.20] the subcomplex  $\text{lk}_K(\sigma)$  is a deformation retract of  $\overline{\text{St}}(\sigma)$ , and in Hatcher's terminology a deformation retraction is by definition a homotopy fixing the subspace pointwise. Choose such a homotopy

$$H^\sigma : \overline{\text{St}}(\sigma) \times I \rightarrow \overline{\text{St}}(\sigma), \quad H_0^\sigma = \text{id}, \quad H_1^\sigma(\overline{\text{St}}(\sigma)) \subseteq \text{lk}_K(\sigma), \quad H_t^\sigma|_{\text{lk}_K(\sigma)} = \text{id}.$$

Define  $H : K \times I \rightarrow K$  by the identity on  $K' \times I$  and by  $H^\sigma$  on  $\overline{\text{St}}(\sigma) \times I$ . The pieces agree on overlaps: by Corollary 6.5 we have  $\overline{\text{St}}(\sigma) \cap K' = \text{lk}_K(\sigma)$ , on which  $H_t^\sigma$  is the identity, and two distinct stars meet inside  $\text{lk}_K(\sigma) \cap \text{lk}_K(\sigma')$ , where both are the identity. So  $H$  is a well-defined map of sets.

For continuity,  $K$  carries the weak topology determined by the family of subcomplexes  $\{K'\} \cup \{\overline{\text{St}}(\sigma)\}_\sigma$ , which is a cover of  $K$  by Proposition 6.7; no local finiteness is needed, closure-finiteness of a CW complex being enough. Since  $I$  is locally compact,  $K \times I$  carries the weak topology determined by  $\{K' \times I\} \cup \{\overline{\text{St}}(\sigma) \times I\}_\sigma$ , by [3, Theorem A.6]. A map out of a space with the weak topology is continuous as soon as its restriction to each member of the family is, and each restriction of  $H$  is continuous. Hence  $H$  is continuous,  $H_0 = \text{id}$ ,  $H_1(K) \subseteq K'$ , and  $H_t|_{K'} = \text{id}$  for all  $t$ .

The last assertion combines this with Proposition 6.1.  $\square$

*Remark 6.9.* There are infinitely many separating vertices, and the point of the pushout is that no ordering of the deletions has to be chosen: by Corollary 6.5 the links are pairwise related only through  $K'$ , so all the homotopies are performed at once. Alternatively one can avoid the simultaneity: a map of a sphere into  $K$  has compact image, hence meets only finitely many separating vertices, so finitely many single-vertex deletions suffice to compute any  $\pi_n$  or  $H_n$ , and Whitehead's theorem then gives the same conclusion.

## 7. SIMPLE CONNECTIVITY

This section proves  $\pi_1(C_1(S_g)) = 1$  for every closed orientable  $S_g$  with  $g \geq 2$  — not only for  $g = 2$ . The argument is independent of the rest of the paper and rests on two inputs: Harer's determination of the homotopy type of  $C_0(S_g)$ , and an elementary surgery on an edge. A second proof, following Souto, is included at the end of the section; it is shorter in spirit but rests on a contractibility statement for a different complex, and we keep the two apart. Two complexes both traditionally written  $A(\cdot)$  appear in this circle of ideas, and they are distinguished here by name and by notation.

**Definition 7.1.** For a compact surface  $S$  and a non-empty finite set  $V \subset S$ , let  $\mathcal{S}(S, V)$  be the set of isotopy classes of essential simple closed curves *and* essential arcs in  $(S, V)$ , an arc being required to meet  $V$  exactly in its two endpoints, which may coincide. Let  $AC(S, V)$  be the complex on  $\mathcal{S}(S, V)$  in which a finite set spans a simplex when its members admit pairwise disjoint representatives, disjointness of arcs being understood in the interiors. We call  $AC(S, V)$  the *arc-and-curve complex*.

Thus  $AC(S, V)$  has more vertices than the arc complex  $A(S, V)$  of Definition 2.5: it admits closed curves as vertices as well as arcs. The two must not be confused, and the letters are kept distinct throughout.

**Fact 7.2** (Hatcher [6, Theorem (a)]). Let  $S$  be a compact surface and  $V \subset S$  a non-empty finite set, whose points may lie in the interior. Then the *arc* complex  $A(S, V)$  is contractible, except when  $S$  is a disc with  $V \subset \partial S$  or an annulus with  $V$  contained in one boundary component, in which cases it is a sphere.

The theorem carries no number in the revised version of [6], where it is stated as Theorem (a); we cite it accordingly. This is the form used in Section 8, where it is applied to  $\mathcal{A} = A(S^2, B)$ .

The following variant, for the arc-and-curve complex, is needed only for Proposition 7.8, which is stated for completeness and is used nowhere in this paper. The main statement of this section, Theorem 7.7, does not use it.

**Fact 7.3.** Let  $S$  be a closed surface of genus  $g \geq 1$  and  $V \subset S$  a non-empty finite set. Then  $AC(S, V)$  is contractible.

*Remark 7.4* (On the provenance of Fact 7.3). This is the form in which Souto [8] states and uses the result, with attribution to [6]. The theorem printed in [6] is about systems of *arcs*; the passage to systems that may also contain closed curves is standard, by the same surgery argument, but it is not the printed statement. We therefore use Fact 7.3 only for Proposition 7.8, and nothing in this paper depends on that proposition. The main line uses only Fact 7.2, and Theorem 7.7 is unconditional.

**Lemma 7.5** (Filling in an edge). *Let  $g \geq 2$  and let  $a, b$  be essential simple closed curves in  $S_g$  with  $i(a, b) = 1$ . Let  $c := \partial N(a \cup b)$ , the boundary of a regular neighbourhood of  $a \cup b$ . Then  $c$  is a single essential simple closed curve,  $i(a, c) = i(b, c) = 0$ , and  $c$  is distinct from  $a$  and from  $b$  as an isotopy class. Consequently  $\{a, b, c\}$  spans a 2-simplex of  $C_1(S_g)$  and  $\{a, c\}, \{b, c\}$  are edges of  $C_0(S_g)$ .*

*Proof.* Since  $a$  and  $b$  are simple and meet in one point transversally, a regular neighbourhood  $N = N(a \cup b)$  is a compact surface with  $\chi(N) = \chi(a) + \chi(b) - \chi(\text{pt}) = 0 + 0 - 1 = -1$ , containing the two curves whose classes generate a rank two subgroup of  $H_1$ ; so  $N$  is a one-holed torus and  $\partial N = c$  is one circle. It is disjoint from  $a$  and  $b$ , which lie in the interior of  $N$ , so  $i(a, c) = i(b, c) = 0$ .

The curve  $c$  is essential: one side of  $c$  is  $N$ , a one-holed torus, and the other side is a compact surface of genus  $g - 1 \geq 1$  with one boundary circle, neither being a disc. Finally  $c \neq a$  and  $c \neq b$  as isotopy classes, since  $i(a, c) = 0$  while  $i(a, b) = 1$ : if  $c$  were isotopic to  $a$  we would get  $i(b, c) = i(b, a) = 1 \neq 0$ . As  $i$  is an invariant of isotopy classes this separates the classes and not merely the representatives.

Pairwise the three classes have  $i \leq 1$ , so they span a 2-simplex of  $C_1(S_g)$ ; the two edges involving  $c$  have  $i = 0$  and so lie in  $C_0(S_g)$ .  $\square$

**Lemma 7.6.** *For  $g \geq 2$  the inclusion  $C_0(S_g) \hookrightarrow C_1(S_g)$  is surjective on  $\pi_1$ .*

*Proof.* Both complexes are connected and have the same vertices, so it suffices to show that every edge path in  $C_1$  is homotopic, with endpoints fixed, to an edge path in  $C_0$ . An edge  $\{a, b\}$  of  $C_1$  has  $i(a, b) \in \{0, 1\}$ . If  $i(a, b) = 0$  it is an edge of  $C_0$ . If  $i(a, b) = 1$ , take  $c$  as in Lemma 7.5: the 2-simplex  $\{a, b, c\}$  of  $C_1$  provides a homotopy rel endpoints from the edge  $\{a, b\}$  to the path  $a \rightarrow c \rightarrow b$ , whose two edges lie in  $C_0$ . Performing this replacement on each edge of a loop with  $i = 1$  gives a loop in  $C_0$  homotopic to it in  $C_1$ .  $\square$

**Theorem 7.7.** *For every  $g \geq 2$  one has  $\pi_1(C_1(S_g)) = 1$ .*

*Proof.* By Harer [7] the complex  $C_0(S_g)$  is homotopy equivalent to a wedge of spheres of dimension  $2g - 2$ . For  $g \geq 2$  one has  $2g - 2 \geq 2$ , and a wedge of spheres of dimension at least two is simply connected; hence  $\pi_1(C_0(S_g)) = 1$ . By Lemma 7.6 the inclusion  $C_0(S_g) \hookrightarrow C_1(S_g)$  is surjective on  $\pi_1$ . A group which is the image of the trivial group is trivial, so  $\pi_1(C_1(S_g)) = 1$ .  $\square$

This proof uses only Harer's theorem and Lemma 7.6; in particular it does not use Fact 7.3, and Theorem 7.7 is therefore unconditional. We record separately Souto's stronger observation, which does use that fact and which is not needed anywhere in this paper.

**Proposition 7.8** (Souto). *Let  $g \geq 2$ . Then the inclusion  $C_0(S_g) \hookrightarrow C_1(S_g)$  is homotopic to a constant map. The proof below uses Fact 7.3.*

*Proof.* Fix a hyperbolic metric on  $S_g$  and identify the vertices of  $C_0(S_g)$  with the simple closed geodesics. There are countably many of these, so we may choose a point  $*$  in  $S_g$  lying on none of them; put  $V = \{*\}$ .

The choice of  $*$  gives an injection  $\mathcal{S}(S_g) \rightarrow \mathcal{S}(S_g, *)$ : a geodesic misses  $*$  and so is a simple closed curve in  $(S_g, *)$ , essential there. If a finite subcollection of  $\mathcal{S}(S_g)$  spans a simplex of  $C_0(S_g)$ , its members are pairwise disjoint, hence span a simplex of  $AC(S_g, *)$ . So the injection extends to a simplicial map

$$u : C_0(S_g) \longrightarrow AC(S_g, *),$$

whose image consists of closed-curve vertices.

In the other direction define  $v_0 : \mathcal{S}(S_g, *) \rightarrow \mathcal{S}(S_g)$  as follows. An element which is a closed curve in  $(S_g, *)$  is, forgetting  $*$ , an essential simple closed curve in  $S_g$ . An element which is an arc has both endpoints at  $*$ , so forgetting  $*$  turns it into a simple closed curve in  $S_g$ , again essential. The composite  $\mathcal{S}(S_g) \rightarrow \mathcal{S}(S_g, *) \rightarrow \mathcal{S}(S_g)$  is the identity.

The map  $v_0$  is not simplicial into  $C_0(S_g)$ : two arcs in  $(S_g, *)$  with disjoint interiors both pass through  $*$ , so the two closed curves they determine meet at  $*$  and need not be disjoint. But they meet only there:

$$\text{a simplex of } AC(S_g, *) \longmapsto \text{classes with pairwise } i \leq 1,$$

because disjoint interiors leave at most the single common point  $*$ . Hence  $v_0$  extends to a simplicial map

$$v : AC(S_g, *) \longrightarrow C_1(S_g),$$

and by construction the composite  $v \circ u$  is the inclusion  $C_0(S_g) \hookrightarrow C_1(S_g)$ .

Now  $S_g$  is closed of genus  $g \geq 2$  and  $V = \{*\}$  is a single interior point, so  $AC(S_g, *)$  is contractible by Fact 7.3. The inclusion  $C_0 \hookrightarrow C_1$  thus factors through a contractible complex and is null-homotopic.  $\square$

Proposition 7.8 gives a second proof of Theorem 7.7, since a null-homotopic map induces the trivial map on  $\pi_1$ ; that route is the one Souto indicates. We have given the unconditional argument first because the second proof rests on Fact 7.3, whose printed source is the arc version.

## 8. THE SPECTRAL SEQUENCE

**8.1. Three empty objects.** Reduced homology of simplicial complexes forces one to distinguish three things which are all called “empty”. We fix notation once and use it without further comment.

- Convention 8.1** (Empty objects). •  $\hat{0}$  denotes the *empty simplex*, the simplex with no vertices. It is a face of every simplex, and we regard it as an element of every non-void simplicial complex; in particular  $\hat{0} \in \mathcal{A}$ .
- $\mathbb{E} := \{\hat{0}\}$  denotes the *empty complex*: the complex whose only simplex is  $\hat{0}$ . It has no vertices, and  $\tilde{H}_{-1}(\mathbb{E}) = \mathbb{Z}$  while  $\tilde{H}_q(\mathbb{E}) = 0$  for  $q \neq -1$ .
  - $\mathbb{O}$  denotes the *void complex*, which contains no simplex at all, not even  $\hat{0}$ . Its augmented chain complex is zero in every degree, so  $\tilde{H}_q(\mathbb{O}) = 0$  for all  $q$ , including  $q = -1$ .

For a complex  $K$  we write  $\tilde{C}_*(K)$  for the augmented chain complex:  $\tilde{C}_n(K)$  is free on the  $n$ -simplices for  $n \geq 0$ ,  $\tilde{C}_{-1}(K) = \mathbb{Z}\langle\hat{0}\rangle$  if  $K \neq \mathbb{O}$  and 0 if  $K = \mathbb{O}$ , and  $\partial_0$  is the augmentation  $\varepsilon$  sending each vertex to  $\hat{0}$ .

The distinction is not pedantry. Below, the bottom stage of the filtration is  $\mathbb{O}$  and not  $\mathbb{E}$ ; reading it as  $\mathbb{E}$  replaces the reduced homology of  $\mathbf{X}$  in the column  $p = 0$  by its unreduced homology, and the conclusion of the paper is then unobtainable. We return to this in Remark 8.8.

## 8.2. Bad vertices and the filtration.

**Definition 8.2.** A *bad face* of a simplex  $\sigma \in \mathcal{A}$  is a face of  $\sigma$  of one of the following two kinds:

- (B1) a single vertex which is a loop class;
- (B2) a pair of distinct non-loop vertices with the same endpoint pair.

The set of *bad vertices* of  $\sigma$  is

$$\text{bv}(\sigma) := \{v \in \sigma : v \text{ lies in some bad face of } \sigma\}.$$

**Lemma 8.3.** If  $\tau \subseteq \sigma$  then  $\text{bv}(\tau) \subseteq \text{bv}(\sigma)$ . Moreover  $\text{bv}(\hat{0}) = \emptyset$ .

*Proof.* A bad face of  $\tau$  is a face of  $\tau$ , hence a face of  $\sigma$ , and it is bad for  $\sigma$  by the same clause of Definition 8.2. The empty simplex has no faces of either kind.  $\square$

**Definition 8.4.** For  $p \geq -1$  put

$$Y_p := \{\sigma \in \mathcal{A} : |\text{bv}(\sigma)| \leq p\},$$

$$\mathcal{S}_p := \{S \in \mathcal{A} : |S| = p, \text{bv}(S) = S\}, \quad L_S := \{\tau \in \text{lk}_{\mathcal{A}}(S) : \text{bv}(S \cup \tau) = S\}.$$

By Lemma 8.3 each  $Y_p$  is a subcomplex.

- Lemma 8.5.** (1)  $Y_{-1} = \mathbb{O}$ , and  $\hat{0} \in Y_0$ .  
 (2)  $\mathcal{S}_0 = \{\hat{0}\}$  and  $L_{\hat{0}} = \mathbf{X}$ .  
 (3)  $Y_0 = \mathbf{X}$ .  
 (4)  $\dim \mathcal{A} = 11$ ,  $Y_{12} = \mathcal{A}$ , and  $\mathcal{S}_p = \emptyset$  for  $p > 12$ .

*Proof.* (i) For every  $\sigma \in \mathcal{A}$  we have  $|\text{bv}(\sigma)| \geq 0$ , and  $0 \leq -1$  is false; so no simplex satisfies the defining condition of  $Y_{-1}$ , not even  $\hat{0}$ , for which  $|\text{bv}(\hat{0})| = 0$  by Lemma 8.3. The same computation puts  $\hat{0}$  in  $Y_0$ .

(ii)  $|S| = 0$  forces  $S = \hat{0}$ , and  $\text{bv}(\hat{0}) = \emptyset = \hat{0}$ , so  $\mathcal{S}_0 = \{\hat{0}\}$ ; this is consistent only because  $\hat{0} \in \mathcal{A}$ , as fixed in Convention 8.1. Next  $\text{lk}_{\mathcal{A}}(\hat{0}) = \mathcal{A}$ , and the condition  $\text{bv}(\hat{0} \cup \tau) = \hat{0}$  reads  $\text{bv}(\tau) = \emptyset$ , that is,  $\tau \in Y_0$ ; so  $L_{\hat{0}} = Y_0$ , which is  $\mathbf{X}$  by (iii).

(iii) A simplex  $\sigma \in \mathcal{A}$  has  $\text{bv}(\sigma) = \emptyset$  if and only if it contains no loop vertex and no two distinct vertices with the same endpoint pair, that is, if and only if its vertices are pairwise adjacent in the graph  $G$  of Definition 2.5. Conversely, let  $\sigma$  be a finite clique of  $G$ . Its members are pairwise realisable with disjoint interiors, so by Theorem 3.13 their geodesic representatives

have pairwise disjoint interiors *simultaneously*; hence  $\sigma$  is a simplex of  $\mathcal{A}$ . Thus the simplices of  $Y_0$  are exactly the cliques of  $G$ , which are the simplices of the flag complex  $X$ .

(iv) A simplex of  $\mathcal{A}$  is a collection of pairwise non-isotopic essential arcs with pairwise disjoint interiors, so its geodesic representatives form an embedded graph in  $S^2$  with vertex set contained in  $B$ , now allowing loops and multiple edges. Such a graph has  $V \leq 6$  and, if all complementary faces are triangles,  $2E = 3F$  and  $V - E + F = 2$ , whence  $E = 3V - 6 \leq 12$ . A maximal collection triangulates, so  $\dim \mathcal{A} = 11$ ; an ideal triangulation of the six-times-marked sphere has 12 arcs and 8 triangles. Finally  $|\text{bv}(\sigma)| \leq |\sigma| \leq 12$  gives  $Y_{12} = \mathcal{A}$ , and  $|S| = p > 12$  is impossible.  $\square$

### 8.3. The splitting.

**Lemma 8.6.** *Let  $p \geq 1$  and let  $\sigma \in \mathcal{A}$  satisfy  $|\text{bv}(\sigma)| = p$ . Then  $\sigma = S \sqcup \tau$  with  $S := \text{bv}(\sigma) \in \mathcal{S}_p$  and  $\tau \in L_S$ , and this decomposition is unique. Moreover:*

- (1) *if  $u \in \tau$  then  $\text{bv}(\sigma \setminus u) = S$ ;*
- (2) *if  $v \in S$  then  $|\text{bv}(\sigma \setminus v)| \leq p - 1$ .*

*Proof.* Put  $S = \text{bv}(\sigma)$  and  $\tau = \sigma \setminus S$ . Then  $|S| = p$ , and  $\text{bv}(S) = S$ : each  $v \in S$  lies in a bad face of  $\sigma$ , and that face is contained in  $S$ , because both (B1) and (B2) consist of vertices which are themselves bad. Also  $\text{bv}(S \cup \tau) = \text{bv}(\sigma) = S$ , so  $\tau \in L_S$ . The decomposition is unique because  $S$  is determined by  $\sigma$ .

(i) By Lemma 8.3,  $\text{bv}(\sigma \setminus u) \subseteq S$ . For the reverse inclusion let  $v \in S$ . If  $v$  is bad by (B1) the bad face is  $\{v\}$ , which survives the removal of  $u \neq v$ . If  $v$  is bad by (B2) there is  $w \neq v$  with  $\{v, w\}$  a bad face; then  $w$  too lies in a bad face, so  $w \in \text{bv}(\sigma) = S$ , whereas  $u \in \tau$  and  $S \cap \tau = \emptyset$ ; hence  $u \neq w$  and the bad face  $\{v, w\}$  survives. In both cases  $v \in \text{bv}(\sigma \setminus u)$ .

(ii) Again  $\text{bv}(\sigma \setminus v) \subseteq S$ , and  $v \notin \sigma \setminus v$ , so  $\text{bv}(\sigma \setminus v) \subseteq S \setminus \{v\}$  has at most  $p - 1$  elements.  $\square$

**Proposition 8.7.** *For  $p \geq 1$  there is an isomorphism of chain complexes*

$$\tilde{C}_n(Y_p, Y_{p-1}) \cong \bigoplus_{S \in \mathcal{S}_p} \tilde{C}_{n-p}(L_S),$$

*where on the right the differential is that of  $\tilde{C}_*(L_S)$  multiplied by  $(-1)^p$ . For  $p = 0$  one has  $\tilde{C}_*(Y_0, Y_{-1}) = \tilde{C}_*(X)$ .*

*Proof.* Let  $p \geq 1$ . The group  $\tilde{C}_n(Y_p, Y_{p-1})$  is free on the simplices  $\sigma$  with  $|\text{bv}(\sigma)| = p$  and  $\dim \sigma = n$ . Decompose  $\sigma = S \sqcup \tau$  as in Lemma 8.6; then  $\dim \tau = n - p$ , with  $\dim \hat{0} = -1$  corresponding to  $\sigma = S$ . Order the vertices of  $\sigma$  so that those of  $S$  come first, in an order fixed once and for all for each  $S \in \mathcal{S}_p$ , followed by those of  $\tau$ ; declare

$$\Phi(\langle s_1, \dots, s_p, t_0, \dots, t_{n-p} \rangle) := \langle t_0, \dots, t_{n-p} \rangle \in \tilde{C}_{n-p}(L_S).$$

This is an isomorphism of graded groups, the  $S$ -blocks being spanned by the  $\sigma$  with  $\text{bv}(\sigma) = S$ .

It intertwines the differentials up to the stated sign. In  $\partial\sigma$ , the terms obtained by deleting a vertex of  $S$  lie in  $Y_{p-1}$  by Lemma 8.6(ii) and so vanish in the relative complex; the terms obtained by deleting a vertex  $t_r$  of  $\tau$  have  $\text{bv} = S$  by Lemma 8.6(i), so they stay in the same  $S$ -block. This is what makes the relative differential block diagonal, and it is the substance of the phrase “the relative complex splits according to  $S$ ”. The surviving term for  $t_r$  carries the sign  $(-1)^{p+r}$ , because  $t_r$  occupies position  $p + r$  in the chosen order, whereas the corresponding term of  $\partial^{L_S} \langle t_0, \dots, t_{n-p} \rangle$  carries  $(-1)^r$ . Hence  $\Phi \circ \partial = (-1)^p \partial^{L_S} \circ \Phi$  on the  $S$ -block. The constant  $(-1)^p$  depends only on  $p$ , so it does not affect kernels, images or homology; note that it cannot be absorbed by rescaling a single generator, and that the two sides must use the *same* order on the vertices of  $\tau$ , which is why the order was fixed in advance.



For  $p = 0$  we have  $Y_{-1} = \mathbb{O}$  by Lemma 8.5(i), so  $\tilde{C}_*(Y_{-1}) = 0$  and the relative complex is  $\tilde{C}_*(Y_0) = \tilde{C}_*(X)$  — augmentation included, since  $\hat{0} \in Y_0$  contributes  $\mathbb{Z}$  in degree  $-1$  and nothing is quotiented away.  $\square$

*Remark 8.8.* Had we put  $Y_{-1} = \mathbb{E}$ , the class of  $\hat{0}$  would have been killed and  $\tilde{C}_*(Y_0, Y_{-1})$  would be the *unreduced* complex  $C_*(X)$ , giving  $E_{0,0}^1 = H_0(X) = \tilde{H}_0(X) \oplus \mathbb{Z}$ . Nothing in the filtration would be false, but the column  $p = 0$  would compute the wrong functor, and the identification of the abutment in Theorem 8.9 would fail. With  $Y_{-1} = \mathbb{O}$  the case  $p = 0$  needs no separate convention: it is the case  $\mathcal{S}_0 = \{\hat{0}\}$ ,  $L_{\hat{0}} = X$  of the general formula.

#### 8.4. The $E^1$ page.

**Theorem 8.9.** *The filtration  $\mathbb{O} = Y_{-1} \subseteq Y_0 \subseteq \cdots \subseteq Y_{12} = \mathcal{A}$  of the augmented chain complex of  $\mathcal{A}$  gives a spectral sequence of a bounded filtration with augmentation — the row  $q = -1$  is present and carries information, so the sequence is not first quadrant — with*

$$E_{p,q}^1 = \bigoplus_{S \in \mathcal{S}_p} \tilde{H}_q(L_S; \mathbb{Z}) \implies \tilde{H}_{p+q}(\mathcal{A}; \mathbb{Z}) = 0, \quad d^r : E_{p,q}^r \rightarrow E_{p-r, q+r-1}^r,$$

valid for all  $p \geq 0$  and  $q \geq -1$ ; in particular  $E_{0,q}^1 = \tilde{H}_q(X; \mathbb{Z})$ . The filtration is bounded, so the sequence converges strongly.

*Proof.* Taking homology of the isomorphism of Proposition 8.7 gives

$$E_{p,q}^1 = H_{p+q}(\tilde{C}_*(Y_p, Y_{p-1})) \cong \bigoplus_{S \in \mathcal{S}_p} \tilde{H}_q(L_S),$$

the sign  $(-1)^p$  being irrelevant to homology; for  $p = 0$  this reads  $E_{0,q}^1 = \tilde{H}_q(X)$ , in agreement with  $\mathcal{S}_0 = \{\hat{0}\}$  and  $L_{\hat{0}} = X$ . The filtration has finite length by Lemma 8.5(iv), so the associated spectral sequence converges strongly to the homology of  $\tilde{C}_*(\mathcal{A})$ . Finally  $\mathcal{A}$  is contractible by Fact 7.2, applied to  $S = S^2$  and  $V = B$ : a sphere with six interior marked points is neither a disc with  $V$  in its boundary nor an annulus. Hence  $\tilde{H}_*(\mathcal{A}; \mathbb{Z}) = 0$ .  $\square$

Note that  $q = -1$  is a genuine part of the range:  $\tilde{H}_{-1}(L_S)$  is  $\mathbb{Z}$  when  $L_S = \mathbb{E}$  and 0 when  $L_S$  has at least one vertex, so the vanishing of the column  $p \geq 1$  will require each  $L_S$  to be non-empty and not merely contractible. This is supplied by Corollary 10.9 below.

### 9. OBJECTS, GAPS AND THE BUDGET

Throughout this section  $p \geq 1$  and  $S \in \mathcal{S}_p$  is fixed. We realise the arcs of  $S$  by their geodesic representatives for the cusped metric and write

$$\Gamma_S := \bigcup S \subset S^2,$$

which by Theorem 3.13 is an embedded graph with vertex set in  $B$ : the representatives have pairwise disjoint interiors simultaneously. The aim of the section is Lemma 9.15, which produces inside  $\Gamma_S$  a complementary region of a very restricted shape.

**9.1. Objects.** The condition  $\text{bv}(S) = S$  says that every vertex of  $S$  is bad, that is, every arc of  $S$  is either a loop or has a partner in  $S$  with the same endpoint pair. This is what the following decomposition exploits.

**Definition 9.1.** For an endpoint pair  $e$  let  $C_e(S)$  be the set of arcs of  $S$  whose endpoint pair is exactly  $e$ . The *objects* of  $S$  are

- each loop of  $S$ , taken singly, and

- each union  $\bigcup C_e(S)$  with  $|C_e(S)| \geq 2$ , taken as one object.

Since  $\text{bv}(S) = S$ , every arc of  $S$  lies in exactly one object, so  $\Gamma_S$  is the union of the objects and the objects are pairwise disjoint except possibly at points of  $B$ .

*Remark 9.2.* The grouping matters. Had we instead grouped arcs into maximal families of pairwise parallel arcs without insisting on a common endpoint pair, two distinct objects could share *two* marked points, and Corollary 9.3, on which the whole section rests, would fail.

**Corollary 9.3.** *Any two distinct objects meet in at most one point of  $B$ . Each object is a connected finite embedded graph in  $S^2$ .*

*Proof.* Two distinct objects of class type have distinct endpoint pairs, and two distinct two-element subsets of  $B$  meet in at most one point. A loop meets any other object at most in its base point, and two distinct loops at most in a common base point. For connectedness: a loop is a circle; an object  $\bigcup C_e(S)$  with  $e = \{a, b\}$  is a union of arcs all joining  $a$  to  $b$ , hence connected.  $\square$

## 9.2. Gaps.

**Definition 9.4.** Let  $O$  be an object. A *gap* of  $O$  is a connected component of  $S^2 \setminus O$ ; the number of gaps is denoted  $\mu(O)$ . A gap  $g$  of  $O$  is *free* if no object other than  $O$  is contained in  $\bar{g}$ .

A *face* of  $\Gamma_S$  is a connected component of  $S^2 \setminus \Gamma_S$ . For a face or gap  $\Delta$  let  $X(\Delta) \subset B$  be the set of marked points on  $\partial\Delta$  and  $k(\Delta)$  the number of marked points in  $\Delta$  itself. Put  $F := \{e : |C_e(S)| \geq 2\}$ . Call  $\Delta$  *clean* if every pair of distinct points of  $X(\Delta)$  belongs to  $F$ , and a *good face* if it is a free gap, is clean, and has  $k(\Delta) \in \{1, 2\}$ .

Finally let  $V'$  be the number of points of  $B$  met by  $\Gamma_S$  and  $n := 6 - V'$ , so that  $n = \sum_{\Delta} k(\Delta)$ , the sum over the faces of  $\Gamma_S$ .

**Lemma 9.5.** *Every gap of every object is an open disc.*

*Proof.* An object is a connected finite graph embedded in  $S^2$  (Corollary 9.3), and an embedding of a connected graph in  $S^2$  is cellular: each complementary component is an open disc. We stress that this is not a consequence of the Euler relation, which holds for every connected plane graph, trees included; it is the statement that a connected compact subset of  $S^2$  whose complement is a union of finitely many components has simply connected complementary components, valid for  $S^2$  and false on surfaces of positive genus.  $\square$

**Lemma 9.6.** *Let  $O$  be an object and  $g$  a gap of  $O$ .*

- (1) *If  $O$  is a loop then  $\mu(O) = 2$ , and  $\partial g$  is the whole circle  $O$ ; the base point occurs exactly once on  $\partial g$ .*
- (2) *If  $O = \bigcup C_e(S)$  consists of  $m \geq 2$  arcs joining  $a \neq b$ , then  $\mu(O) = m$ , each gap is bounded by exactly two of the arcs,  $\partial g$  is a union of two arcs, and  $a$  and  $b$  each occur exactly once on  $\partial g$ .*

*In both cases  $\partial g$  has the cardinality of the continuum; in particular it is not finite.*

*Proof.* (i) is the Jordan curve theorem.

(ii) The object is a plane graph with  $V = 2$ ,  $E = m$  and no loop edges, so  $F = 2 - V + E = m$  by Euler's relation, which gives the number of gaps. The sum of the face degrees is  $2E = 2m$ . No edge is a loop edge, so every face has degree at least 2; with  $m$  faces and degrees summing to  $2m$ , every face has degree exactly 2. Thus each gap is bounded by two of the arcs and its boundary walk is  $a \rightarrow b \rightarrow a$ , visiting each of  $a, b$  once.

In case (i)  $\partial g$  is a circle and in case (ii) a union of two arcs; both have cardinality  $2^{\aleph_0}$ .  $\square$

**Corollary 9.7.** *Let  $O \neq O'$  be objects,  $g$  a gap of  $O$  and  $h$  a gap of  $O'$ . Then  $g \neq h$ .*

*Proof.* Suppose  $g = h$ . A gap is a component of the complement of the object, so  $\partial g \subseteq O$  and  $\partial h \subseteq O'$ : indeed if  $x \in \partial g$  and  $x \notin O$ , then  $x$  lies in some component  $g''$  of the open set  $S^2 \setminus O$ ; as  $g''$  is open and every neighbourhood of  $x$  meets  $g$ , we get  $g'' = g$  and  $x \in g$ , contradicting that  $g$  is open and disjoint from its boundary. Hence  $\partial g = \partial h \subseteq O \cap O'$ , which has at most one point by Corollary 9.3, contradicting Lemma 9.6.  $\square$

**Lemma 9.8.** *Let  $O \neq O'$  be objects and let  $p_0$  be the point of  $O \cap O'$  if there is one. Then there is a unique gap  $g$  of  $O$  with*

$$O' \subseteq g \cup \{p_0\} \subseteq \bar{g}$$

(and  $O' \subseteq g$  if  $O \cap O' = \emptyset$ ).

*Proof.* By Corollary 9.3,  $O \cap O' \subseteq \{p_0\}$ . The set  $O' \setminus \{p_0\}$  is connected: a loop with its base point removed is an open arc, and  $\bigcup C_e(S)$  with one endpoint removed is still joined at the other endpoint. Being a connected subset of  $S^2 \setminus O$ , it lies in a unique gap  $g$ . Then  $O' \subseteq g \cup \{p_0\}$ , and  $p_0$  is a limit point of  $O' \setminus \{p_0\} \subseteq g$ , so  $p_0 \in \bar{g}$ .  $\square$

**Lemma 9.9.** *Let  $g$  be a gap of an object  $O$  which is not free. Then  $g$  properly contains some gap; in particular  $g$  is not minimal with respect to inclusion among all gaps of all objects.*

*Proof.* Let  $O' \neq O$  be an object with  $O' \subseteq \bar{g}$ . Applying Lemma 9.8 with the roles exchanged, let  $h'$  be the gap of  $O'$  with  $O \subseteq \bar{h}'$ , and, using  $\mu(O') \geq 2$ , pick another gap  $h \neq h'$  of  $O'$ .

*Step 1:*  $h \cap O = \emptyset$ . As  $h, h'$  are distinct components of  $S^2 \setminus O'$  they are disjoint, so

$$h \cap \bar{h}' \subseteq h \cap \partial h' \subseteq h \cap O' = \emptyset,$$

and  $O \subseteq \bar{h}'$  gives  $h \cap O = \emptyset$ .

*Step 2:*  $h \subseteq g$ . By Step 1 and connectedness,  $h$  lies in a single gap of  $O$ . Also  $\partial h \subseteq O'$ , while  $O' \subseteq \bar{g} = g \cup \partial g$  with  $\partial g \subseteq O$  and  $g \cap O = \emptyset$ ; hence  $\bar{g} \setminus O = g$ , and with  $O \cap O' \subseteq \{p_0\}$ ,

$$\partial h \setminus \{p_0\} \subseteq O' \setminus O \subseteq \bar{g} \setminus O = g.$$

By Lemma 9.6 the set  $\partial h$  is infinite, so  $\partial h \setminus \{p_0\} \neq \emptyset$ ; pick  $y$  in it. Then  $y \in g$  and  $g$  is open, so a neighbourhood  $U \subseteq g$  of  $y$  meets  $h$ , because  $y \in \partial h$ . Hence  $h \cap g \neq \emptyset$  and the gap of  $O$  containing  $h$  is  $g$ .

*Step 3: properness.*  $O' \setminus \{p_0\} \subseteq O' \setminus O \subseteq g$  is non-empty and disjoint from  $h$ , since  $h \cap O' = \emptyset$ . So  $h \subsetneq g$ .  $\square$

The contrapositive will be used repeatedly: *a minimal gap is free.*

**Lemma 9.10.** *Let  $g$  be a minimal gap of an object  $O$  and let  $h$  be a gap of an object  $O' \neq O$ . Then  $g \subseteq h$  or  $g \cap h = \emptyset$ . If  $h$  is minimal as well, then  $g \cap h = \emptyset$ .*

*Proof.* By Lemma 9.9 the gap  $g$  is free, so no object lies in  $\bar{g}$ ; in particular  $O' \not\subseteq \bar{g}$ . By Lemma 9.8 there is a gap  $g'$  of  $O$  with  $O' \subseteq \bar{g}'$ , and necessarily  $g' \neq g$ . Therefore

$$O' \cap g \subseteq \bar{g}' \cap g = (g' \cup \partial g') \cap g \subseteq (g' \cap g) \cup (O \cap g) = \emptyset,$$

because distinct gaps of the same object are distinct components of  $S^2 \setminus O$ , and  $\partial g' \subseteq O$  while  $g \cap O = \emptyset$ . Thus  $g$  is a connected set disjoint from  $O'$ , hence contained in a single gap of  $O'$ ; that gap is either  $h$ , giving  $g \subseteq h$ , or another one, giving  $g \cap h = \emptyset$ .

Now let  $h$  be minimal too and suppose  $g \subseteq h$ . If  $g \neq h$  then  $g \subsetneq h$  contradicts the minimality of  $h$ ; and  $g = h$  is impossible by Corollary 9.7. Hence  $g \cap h = \emptyset$ .  $\square$

**Lemma 9.11.**  *$S$  has at least two disjoint free gaps, and every free gap is a face of  $\Gamma_S$ .*

*Proof. Finiteness.* There are finitely many objects and each has finitely many gaps, so the set of all gaps is finite.

*Every gap contains a minimal one.* If  $g$  is not minimal it properly contains a gap; by finiteness a strictly descending chain terminates, and its last term is minimal.

*Two disjoint free gaps.* Since  $p \geq 1$  there is an object  $O$ , and  $\mu(O) \geq 2$  by Lemma 9.6. Pick two gaps  $g_1 \neq g_2$  of  $O$ ; they are disjoint. Choose minimal gaps  $f_1 \subseteq g_1$  and  $f_2 \subseteq g_2$ ; then  $f_1 \cap f_2 = \emptyset$ , so  $f_1 \neq f_2$ , and both are free by Lemma 9.9.

*Free gaps are faces.* Let  $f$  be a free gap of an object  $O''$ . Every other object  $O'''$  satisfies  $O''' \subseteq g \cup \{p_0\}$  for some gap  $g$  of  $O''$  by Lemma 9.8; freeness forces  $g \neq f$ , and  $p_0 \in O''$  so  $p_0 \notin f$ ; hence  $O''' \cap f = \emptyset$ . Together with  $f \cap O'' = \emptyset$  this gives  $f \cap \Gamma_S = \emptyset$ . Now  $f$  is open and connected with  $\partial f \subseteq O'' \subseteq \Gamma_S$ . Let  $W$  be the component of  $S^2 \setminus \Gamma_S$  containing  $f$ ; it is open, hence locally path connected, hence path connected. If  $W \supsetneq f$  there is a path in  $W$  from a point of  $f$  to a point of  $W \setminus f$ ; letting  $t^*$  be the supremum of the times up to which the path stays in  $f$ , the point at time  $t^*$  lies in  $\bar{f} \setminus f = \partial f \subseteq \Gamma_S$ , contradicting  $W \cap \Gamma_S = \emptyset$ . So  $f = W$  is a face.  $\square$

**Corollary 9.12.** *Let  $\Delta$  be a free gap, of the object  $O''$ . Then  $\partial\Delta \subseteq O''$  and:*

- if  $O''$  is a loop,  $X(\Delta)$  is the single base point;
- if  $O'' = \bigcup C_e(S)$ , then  $X(\Delta) = e$  and  $\Delta$  is bounded by exactly two arcs of  $C_e(S)$ .

*In both cases  $|X(\Delta)| \leq 2$  and each point of  $X(\Delta)$  occurs exactly once on  $\partial\Delta$ .*

*Proof.* Combine the last part of Lemma 9.11 with Lemma 9.6.  $\square$

**Corollary 9.13.** *Every free gap is clean.*

*Proof.* If  $O''$  is a loop then  $|X(\Delta)| = 1$  and the condition on pairs is vacuous. If  $O'' = \bigcup C_e(S)$  then  $X(\Delta) = e$ , whose only pair is  $e$  itself, and  $e \in F$  because  $|C_e(S)| \geq 2$ .  $\square$

### 9.3. Every free gap uses a marked point.

**Lemma 9.14.** *Every free gap  $g$  satisfies  $k(g) \geq 1$ .*

*Proof.* Let  $g$  be a free gap of the object  $O''$  and suppose  $k(g) = 0$ .

If  $O''$  is a loop  $\ell$ , then by Corollary 9.12 the closure  $\bar{g}$  is a closed disc bounded by  $\ell$  whose interior contains no marked point, and  $\partial\bar{g} = \ell$  meets  $B$  only at the base point. So  $\ell$  separates  $S^2$  into two pieces one of which is a disc meeting  $B$  only in the endpoints of  $\ell$ ; by Definition 2.3 the arc  $\ell$  is not essential, a contradiction.

If  $O'' = \bigcup C_e(S)$ , then by Corollary 9.12 we have  $\partial g = \beta \cup \beta'$  with  $\beta \neq \beta'$  two arcs of  $C_e(S)$ . Then  $g$  is an open disc bounded by  $\beta \cup \beta'$  and containing no marked point, so  $\beta$  and  $\beta'$  are isotopic rel endpoints, and a fortiori isotopic in the sense of Convention 2.1. Hence they are the same vertex of  $\mathcal{A}$ , contradicting  $\beta \neq \beta'$ .  $\square$

### 9.4. The budget.

**Lemma 9.15** (Budget).  $\Gamma_S$  has a good face.

*Proof.* By Lemma 9.11 choose two disjoint free gaps  $g_1 \neq g_2$ ; by Corollary 9.13 they are clean, by Lemma 9.11 they are faces of  $\Gamma_S$ , and by Lemma 9.14 each has  $k \geq 1$ . Being disjoint faces, the marked points they contain are distinct, so

$$k(g_1) + k(g_2) \leq \sum_{\Delta \text{ face}} k(\Delta) = n = 6 - V'.$$

As  $S \neq \hat{0}$  we have  $V' \geq 1$ , hence  $n \leq 5$ . If both  $k(g_m) \geq 3$  then the left side is at least  $6 > 5$ . So some  $g_m$  has  $k(g_m) \in \{1, 2\}$ , and that  $g_m$  is a good face.  $\square$

**Corollary 9.16.**  *$n \geq 2$ , that is,  $V' \leq 4$ : a stratum never meets five or six of the marked points.*

*Proof.* The minimal gaps are pairwise disjoint by Lemma 9.10, each is free by Lemma 9.9 and hence a face, and each contains a marked point of its own by Lemma 9.14. There are at least two of them, as in the proof of Lemma 9.11. Hence  $n \geq 2$ .  $\square$

The bound  $V' \leq 4$  is sharp: take  $a_1, a_2$  joining 1 to 2, with the two sides of  $a_1 \cup a_2$  containing 6 and  $\{3, 4, 5\}$  respectively, and inside the second side  $b_1, b_2$  joining 3 to 4 with sides containing 5 and  $\{1, 2, 6\}$ . All four arcs are essential, the two classes  $C_{12}$  and  $C_{34}$  are the objects, and  $V' = 4$ .

## 10. THE LINK COMPLEXES ARE CONES

We keep  $p \geq 1$  and  $S \in \mathcal{S}_p$  fixed.

### 10.1. $L_S$ is a full flag subcomplex.

**Lemma 10.1.** *Put*

$$V(L_S) := \{v \in \mathbf{X} : v \cap \Gamma_S \subseteq B, \text{ and } \text{ep}(v) \neq \text{ep}(\beta) \text{ for every non-loop } \beta \in S\}.$$

*Then  $L_S$  is the full subcomplex of  $\mathbf{X}$  on  $V(L_S)$ ; in particular  $L_S$  is a flag complex. Moreover the interior of each  $v \in V(L_S)$  lies in a single face of  $\Gamma_S$ .*

*Proof.* For  $\tau \in \text{lk}_{\mathcal{A}}(S)$  the condition  $\text{bv}(S \cup \tau) = S$  says three things:  $\tau$  contains no loop;  $\tau$  contains no two vertices with a common endpoint pair; and no vertex of  $\tau$  has the endpoint pair of a non-loop vertex of  $S$ . The second is automatic in  $\mathbf{X}$ , where two arcs with the same endpoint pair are non-adjacent and so never lie in a common simplex. The first and the third, together with  $\tau \in \text{lk}_{\mathcal{A}}(S)$ , are conditions on each vertex separately. Hence membership in  $L_S$  is determined by the vertex set, that is,  $L_S$  is a full subcomplex of  $\mathbf{X}$ ; a full subcomplex of a flag complex is flag.

For the last statement, let  $v \in V(L_S)$ . Realise  $v$  and the arcs of  $S$  by their geodesic representatives. By Theorem 3.13 these realise all pairwise minima simultaneously, and  $v$  is realisable with interior disjoint from each arc of  $S$ , so the geodesic representative of  $v$  has interior disjoint from  $\Gamma_S$ . Being connected, that interior lies in a single component of  $S^2 \setminus \Gamma_S$ .  $\square$

For a face  $\Delta$  of  $\Gamma_S$  let  $\mathbf{X}_F(\Delta)$  denote the full subcomplex of  $L_S$  on the vertices whose interior lies in  $\Delta$ .

### 10.2. Two uniqueness lemmas.

**Lemma 10.2.** *Let  $\Delta$  be a free gap with  $k(\Delta) = 1$  and interior marked point  $q$ . For each  $x \in X(\Delta)$  there is exactly one class, in the sense of Convention 2.1, of arc from  $x$  to  $q$  contained in  $\Delta$ .*

*Proof.* The set  $\Delta \setminus \{q\}$  is an annulus, and the classes of arcs from  $x$  to  $q$  in  $\Delta$  taken rel  $\partial\Delta$  form a  $\mathbb{Z}$ -torsor, two consecutive classes differing by the Dehn twist  $T_c$  about the core  $c$  of that annulus. The torsor statement is the standard computation that the mapping class group of an annulus rel its boundary is infinite cyclic, generated by the twist about the core; we cite it here rather than leave it implicit, since it is what makes the indexing of the classes by  $\mathbb{Z}$  legitimate. Now  $c$  bounds a disc in  $S^2$  containing exactly one point of  $B$ , namely  $q$ , and a Dehn twist about a curve bounding a disc with at most one marked point is trivial in  $\text{Mod}(S^2, B)$ ; see Farb–Margalit

[2, §3.5], where this is stated for inessential curves in a marked surface. Hence all these classes coincide under Convention 2.1.

Here it is used that  $x$  occurs exactly once on  $\partial\Delta$ , which holds by Corollary 9.12; otherwise two arcs from the two occurrences of  $x$  to  $q$  would be non-isotopic and would have the same endpoint pair.  $\square$

*Remark 10.3.* The lemma is false if isotopies are taken rel  $\partial\Delta$ , so Convention 2.1 is not a matter of taste here. If the rel  $\partial\Delta$  classes were not collapsed, the arcs from  $x$  to  $q$  would give infinitely many vertices of  $\mathbf{X}$  with the same endpoint pair, hence pairwise non-adjacent by (B2), and no such arc could serve as an apex below.

**Lemma 10.4.** *Let  $\Delta$  be a free gap with  $k(\Delta) = 2$  and interior marked points  $q_1, q_2$ . Then all arcs in  $\Delta$  joining  $q_1$  to  $q_2$  are isotopic in the sense of Convention 2.1. We write  $r$  for the resulting class; note that the statement is unordered in  $q_1, q_2$ .*

*Proof.*  $\Delta$  is an open disc, so  $\text{int } \Delta \cong \mathbb{R}^2$ , and the only marked points of  $B$  inside it are  $q_1, q_2$ . By the fact recalled in the proof of Lemma 4.8 — Epstein’s isotopy theorem [10] in the arc case rel endpoints — any two embedded arcs in  $\mathbb{R}^2$  joining two given points are carried onto each other by an ambient isotopy with compact support. Such an isotopy is supported in a compact subset of  $\text{int } \Delta$ , hence fixes  $\partial\Delta$  and every point of  $B$  other than  $q_1, q_2$ , and it fixes  $q_1, q_2$  because they are the endpoints. It is therefore admissible for Convention 2.1.  $\square$

**10.3. The apex on a good face.** Two facts about a good face  $\Delta$  are used, and both come from freeness. First, by Corollary 9.12, each point of  $X(\Delta)$  occurs exactly once on  $\partial\Delta$ . Second:

**Lemma 10.5.** *Let  $\Delta$  be a free gap. Then  $\mathbf{X}_F(\Delta)$  contains no vertex both of whose endpoints lie on  $\partial\Delta$ .*

*Proof.* By Corollary 9.12 the boundary of  $\Delta$  carries the marked points of a single object. If that object is a loop,  $X(\Delta)$  is a single point  $p_0$  and such an arc would be a loop at  $p_0$ , hence not a vertex of  $\mathbf{X}$  by (B1). If the object is  $\bigcup C_e(S)$ , then  $X(\Delta) = e$  and such an arc would have endpoint pair exactly  $e$ , hence the endpoint pair of a member of  $S$ ; it is excluded from  $V(L_S)$  by Lemma 10.1.  $\square$

This is the content of the cleanliness condition, and it is needed: an arc with both endpoints on  $\partial\Delta$  can separate  $q_1$  from  $q_2$  inside  $\Delta$ , and would then meet  $r$ .

**Lemma 10.6.** *Let  $\Delta$  be a good face. Then  $\mathbf{X}_F(\Delta)$  is non-empty and has a vertex  $v_0$  adjacent in  $\mathbf{X}$  to every other vertex of  $\mathbf{X}_F(\Delta)$ ; moreover  $v_0$  has at least one endpoint at an interior marked point of  $\Delta$ .*

*Proof.* By Lemma 10.5 every vertex of  $\mathbf{X}_F(\Delta)$  has at least one endpoint among the interior marked points of  $\Delta$ .

*Case  $k(\Delta) = 1$ , with interior point  $q$ .* Then every vertex of  $\mathbf{X}_F(\Delta)$  joins some  $x \in X(\Delta)$  to  $q$ , and by Lemma 10.2 there is exactly one such vertex for each  $x$ . These can be realised simultaneously as a star centred at  $q$ , so they have pairwise disjoint interiors; their endpoint pairs  $\{x, q\}$  are pairwise distinct because the  $x$  are. Hence  $\mathbf{X}_F(\Delta)$  is a simplex on  $|X(\Delta)|$  vertices, which is non-empty since  $X(\Delta) \neq \emptyset$ , and any vertex may serve as  $v_0$ .

*Case  $k(\Delta) = 2$ , with interior points  $q_1, q_2$ .* The vertices of  $\mathbf{X}_F(\Delta)$  are the classes  $x-q_i$  together with the class  $r$  of Lemma 10.4. Take  $v_0 := r$ , whose two endpoints are both interior marked points. Let  $w$  be a vertex of  $\mathbf{X}_F(\Delta)$  joining  $x \in X(\Delta)$  to  $q_i$ ; we construct a representative of  $r$  meeting  $w$  only at  $q_i$ .



Work in  $\text{int } \Delta \cong \mathbb{R}^2$ . By the same fact ([10]; see the proof of Lemma 4.8) we may choose coordinates in which  $w$  is a straight segment near  $q_i$ . Choose  $\varepsilon > 0$  so small that the closed disc  $N$  of radius  $\varepsilon$  about  $q_i$  lies in  $\Delta$ , meets  $w$  in a single radial segment, and contains neither  $q_{3-i}$  nor any point of  $B$  other than  $q_i$ . Let  $\varrho : [0, 1] \rightarrow N$  be the radial segment from  $\varrho(0) = q_i$  to a point  $P_0 \in \partial N$  in the direction *opposite* to that of  $w$ ; then  $\varrho((0, 1]) \cap w = \emptyset$ , so  $\varrho \cap w = \{q_i\}$ .

Next,  $\Delta \setminus w$  is connected:  $\Delta$  is an open disc and  $w \cap \Delta$  is a closed arc with one endpoint  $q_i$  in  $\Delta$  and its other end approaching  $\partial\Delta$ , so  $\Delta \setminus w$  is an open disc slit along an arc reaching the boundary, which is connected. It is also open, hence path connected, and any two of its points are joined by an embedded arc in it. Both  $P_0$  and  $q_{3-i}$  lie in  $\Delta \setminus w$ , so choose an embedded arc  $\gamma : [0, 1] \rightarrow \Delta \setminus w$  with  $\gamma(0) = q_{3-i}$  and  $\gamma(1) = P_0$ .

Let

$$t^* := \min\{t \in [0, 1] : \gamma(t) \in \varrho([0, 1])\},$$

which exists because the set is closed and contains  $t = 1$ . Then  $\gamma([0, t^*)) \cap \varrho([0, 1]) = \emptyset$ , and  $P := \gamma(t^*) = \varrho(s^*)$  for a unique  $s^*$ . Put

$$r' := \varrho([0, s^*]) \cup \gamma([0, t^*]).$$

The two pieces are embedded arcs meeting exactly in the single point  $P$ , which is an endpoint of each, so  $r'$  is an embedded arc; its endpoints are  $q_i$  and  $q_{3-i}$ , and its interior avoids  $B$ . Finally

$$r' \cap w = (\varrho([0, s^*]) \cap w) \cup (\gamma([0, t^*]) \cap w) = \{q_i\} \cup \emptyset = \{q_i\}.$$

Thus  $r'$  and  $w$  have disjoint interiors. Their endpoint pairs differ, since  $\text{ep}(r') = \{q_1, q_2\}$  consists of interior marked points while  $x \in X(\Delta)$  lies on  $\partial\Delta$ . So  $r'$  and  $w$  are adjacent in  $\mathbf{X}$ , and by Lemma 10.4 the arc  $r'$  represents  $r$ . Hence  $r$  is adjacent to  $w$ .  $\square$

*Remark 10.7.* The argument in the case  $k = 2$  uses neither a change of coordinates principle nor a cutting lemma: no auxiliary homeomorphism of  $\Delta$  is produced, and in particular nothing has to be said about whether such a homeomorphism fixes  $\partial\Delta$  pointwise. It also does not use that a point of  $X(\Delta)$  occurs only once on  $\partial\Delta$ ; that hypothesis is needed only in the case  $k = 1$ , through Lemma 10.2.

#### 10.4. The cone lemma.

**Lemma 10.8** (Cone lemma). *If  $\Gamma_S$  has a good face  $\Delta$ , then  $L_S$  is a cone with apex the vertex  $v_0$  of Lemma 10.6. In particular  $L_S$  is non-empty and contractible, and  $\tilde{H}_q(L_S; \mathbb{Z}) = 0$  for every  $q \geq -1$ .*

*Proof.* First,  $v_0 \in V(L_S)$ : its interior lies in  $\Delta$ , which is disjoint from  $\Gamma_S$ , and one of its endpoints is an interior marked point  $q$  of  $\Delta$ , which is not met by  $\Gamma_S$ ; so the endpoint pair of  $v_0$  is not the endpoint pair of any arc of  $S$ .

Let  $w \in V(L_S)$  with  $w \neq v_0$ . By Lemma 10.1 the interior of  $w$  lies in a single face  $\Delta'$  of  $\Gamma_S$ .

If  $\Delta' \neq \Delta$ , then  $\Delta$  and  $\Delta'$  are disjoint open sets, so  $w$  and  $v_0$  have disjoint interiors and can share at most a marked point on  $\partial\Delta \cap \partial\Delta'$ . Their endpoint pairs differ, because the interior marked point  $q$  of  $\Delta$  is an endpoint of  $v_0$  but cannot be an endpoint of  $w$ , lying in  $\Delta$  and not on its boundary. So  $w$  and  $v_0$  are adjacent.

If  $\Delta' = \Delta$  then  $w \in \mathbf{X}_F(\Delta)$  and  $w$  is adjacent to  $v_0$  by Lemma 10.6.

So  $v_0$  is adjacent to every other vertex of  $L_S$ . As  $L_S$  is flag by Lemma 10.1, a vertex adjacent to all others makes it a cone: any simplex together with  $v_0$  spans a simplex. A cone is contractible, and a non-empty complex has  $\tilde{H}_{-1} = 0$ .  $\square$

**Corollary 10.9.**  *$L_S \neq \emptyset$  for every  $S \in \mathcal{S}_p$  and every  $p$ ; consequently  $E_{p,-1}^1 = 0$  for every  $p \geq 1$ .*

*Proof.* For  $p \geq 1$  combine Lemma 9.15 with Lemma 10.8. For  $p = 0$  we have  $L_{\hat{0}} = X$ , which has vertices.  $\square$

## 11. PROOF OF THE MAIN THEOREM

**Theorem 11.1.**  $E_{p,q}^1 = 0$  for every  $p \geq 1$  and every  $q \geq -1$ .

*Proof.* Fix  $S \in \mathcal{S}_p$  with  $p \geq 1$ . By Lemma 9.15 the graph  $\Gamma_S$  has a good face, so by Lemma 10.8 the complex  $L_S$  is a non-empty cone and  $\tilde{H}_q(L_S) = 0$  for all  $q \geq -1$ ; the case  $q = -1$  is exactly the non-emptiness, recorded in Corollary 10.9. Summing over  $\mathcal{S}_p$  and using Theorem 8.9 gives  $E_{p,q}^1 = 0$ .  $\square$

**Theorem 11.2.**  $\tilde{H}_*(X; \mathbb{Z}) = 0$  and  $\tilde{H}_*(C_1(S_2); \mathbb{Z}) = 0$ .

*Proof.* By Theorem 11.1 the page  $E^1$  is concentrated in the single column  $p = 0$ . Every differential  $d^r$  with  $r \geq 1$  has source and target in different columns, since it lowers  $p$  by  $r \geq 1$ ; hence all differentials vanish and  $E^1 = E^\infty$ . The filtration being bounded, Theorem 8.9 gives that  $E_{0,q}^\infty$  is a subquotient of the associated graded of  $\tilde{H}_q(\mathcal{A}; \mathbb{Z}) = 0$ , so  $E_{0,q}^\infty = 0$ . Therefore

$$\tilde{H}_q(X; \mathbb{Z}) = E_{0,q}^1 = E_{0,q}^\infty = 0 \quad \text{for every } q.$$

By Theorem 6.8 the complexes  $C_1(S_2)$  and  $X$  are homotopy equivalent, so the reduced homology of  $C_1(S_2)$  vanishes as well.  $\square$

*Proof of Theorem 1.1.* The complex  $C_1(S_2)$  is a simplicial, hence CW, complex. It is simply connected by Theorem 7.7 applied with  $g = 2$ , and all its reduced integral homology vanishes by Theorem 11.2. A simply connected space with vanishing reduced integral homology has vanishing homotopy groups in all degrees, by the Hurewicz theorem applied inductively. A CW complex all of whose homotopy groups vanish is contractible, by Whitehead's theorem applied to the map to a point. Hence  $C_1(S_2)$  is contractible.  $\square$

**Corollary 11.3.**  $H_k(C_1(S_2); \mathbb{Z}) = 0$  for every  $k \geq 1$  and  $H_0(C_1(S_2); \mathbb{Z}) = \mathbb{Z}$ . The non-separating subcomplex  $C_1^{\text{ns}}(S_2) \cong X$  is contractible as well.

*Proof.* The first statement is Theorem 11.2. For the second,  $X \simeq C_1(S_2)$  by Theorem 6.8 and a complex homotopy equivalent to a contractible complex is contractible.  $\square$

**11.1. Where the argument stops.** It may be worth isolating the three points at which the genus enters, since each of them blocks a naive extension.

First, the number  $6 = 2g + 2$  of branch points is the budget of Lemma 9.15: the inequality  $k(g_1) + k(g_2) \leq 6 - V' \leq 5$  is what forces one of the two free gaps to carry one or two marked points. For  $g = 3$  the same scheme gives eight branch points and a budget of 7, and two free gaps with  $k = 3$  and  $k = 4$  are no longer excluded, so the good face need not exist.

Second, and more decisively, a closed surface of genus  $g \geq 3$  has in general no hyperelliptic involution, so there is no quotient marked sphere and the dictionary of Section 4 has no analogue. The whole method is therefore confined to  $g = 2$ ; Theorem 1.1 gives no information for  $g \geq 3$ .

Third, nothing is proved here about punctured or bordered surfaces. The natural simplicial map forgets the puncture and runs from  $C_1(\Sigma_{g,1})$  to  $C_1(S_g)$ ; it is surjective on vertices and not injective. Whether the two complexes are homotopy equivalent is left open, and no implication between the punctured and the closed case is claimed in either direction. We flag this because an argument which assumes the closed case in order to derive the punctured case, and then recovers

the closed case from it, would be circular, and that trap is easy to fall into if the two complexes are treated as interchangeable.

*Remark 11.4.* No step of this paper computes homology of a finite subcomplex. This is deliberate: for the complexes considered here there is in general no map, in either direction, between the homology of a finite pool of vertices, or of a finite quotient poset, and the homology of the ambient complex, so such a computation could not be evidence for Theorem 1.1. Every argument above is structural.

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